



Context-Aware GenAI for the Enterprise: RAG, Agent Design & Modular Orchestration

YVES SCHILLINGS

Why AI Agents, RAG & Orchestration Matter for Enterprise Intelligence

Phase 1 Foundational Enablement	1.	Understand the limitations of standalone Large Languages Models (LLMs)
	2.	Understand how vector stores and embeddings power semantic retrieval
	3.	Follow the Retrieval-Augmented Generation (RAG) pipeline
	4.	How the Model Context Protocol (MCP) structures context for the LLM
	5.	Context-Aware Generation (CAG)
Phase 2 Multi-Agent Integration	6.	Learn how agents use LLMs to reason and act autonomously
	7.	Multi-Agentic Workflow with LangGraph and Shared Context
Phase 3 Enterprise-Scale AI Ecosystem	8.	Security & Compliance in Context-Aware GenAI Architectures
	9.	Monitoring, Tracing & Observability in Azure
	10.	Takeaways: Strategic Impact of AI Agents for the Enterprise

Phase 1 – Foundational Enablement

Establish the core technical and functional building blocks to execute an initial use case

From Prompt Experiments to Enterprise-Grade AI

The Problem Today

- Most **enterprise AI pilots fail** to move beyond prototypes
- **LLM outputs** are often **inaccurate, ungoverned, and disconnected from business-critical knowledge**
- **Compliance** and **explainability** are afterthoughts

What's Changing

Retrieval-Augmented Generation (RAG)

Model Context Protocol (MCP)

Autonomous agents

⇒ create governed, explainable, and scalable AI systems

- **Context-aware pipelines** inject the right data into the LLM at the right time.
- **Autonomous agents turn answers into decisions and actions**

What You'll Learn Today

- How to **ground LLMs** in your **enterprise knowledge base**
- How to **orchestrate AI workflows** with **MCP** and **modular agents**
- How to deliver **production-ready, compliant AI** at scale

1. Understand the limitations of standalone LLMs

Stateless, hallucination-prone, and disconnected from enterprise knowledge



The problem with static Large Language Models (LLMs)

Why Prompting alone is not enough

Limitations of LLMs in Enterprise Contexts

- Stateless: **no memory of prior interactions**
⇒ costly and fragmented.
- Disconnected: can't access real-time or internal data (ex: policies, CRM, procedures)

Prompting Alone Lacks

- Memory: **no persistent context** ⇒ frustrating UX
- Grounding: **hallucinations due to lack of domain-specific data**
- Domain Reasoning: **struggles with complex or multi-step tasks**

Consequences

- Hallucination ⇒ Risk of misinformation
- Generic answers ⇒ Lack of business value
- Inconsistent quality ⇒ Unpredictable UX

Solution

- Combine LLMs with **Retrieval-Augmented Generation (RAG)**
- Inject live, **contextual knowledge**, and reasoning capability

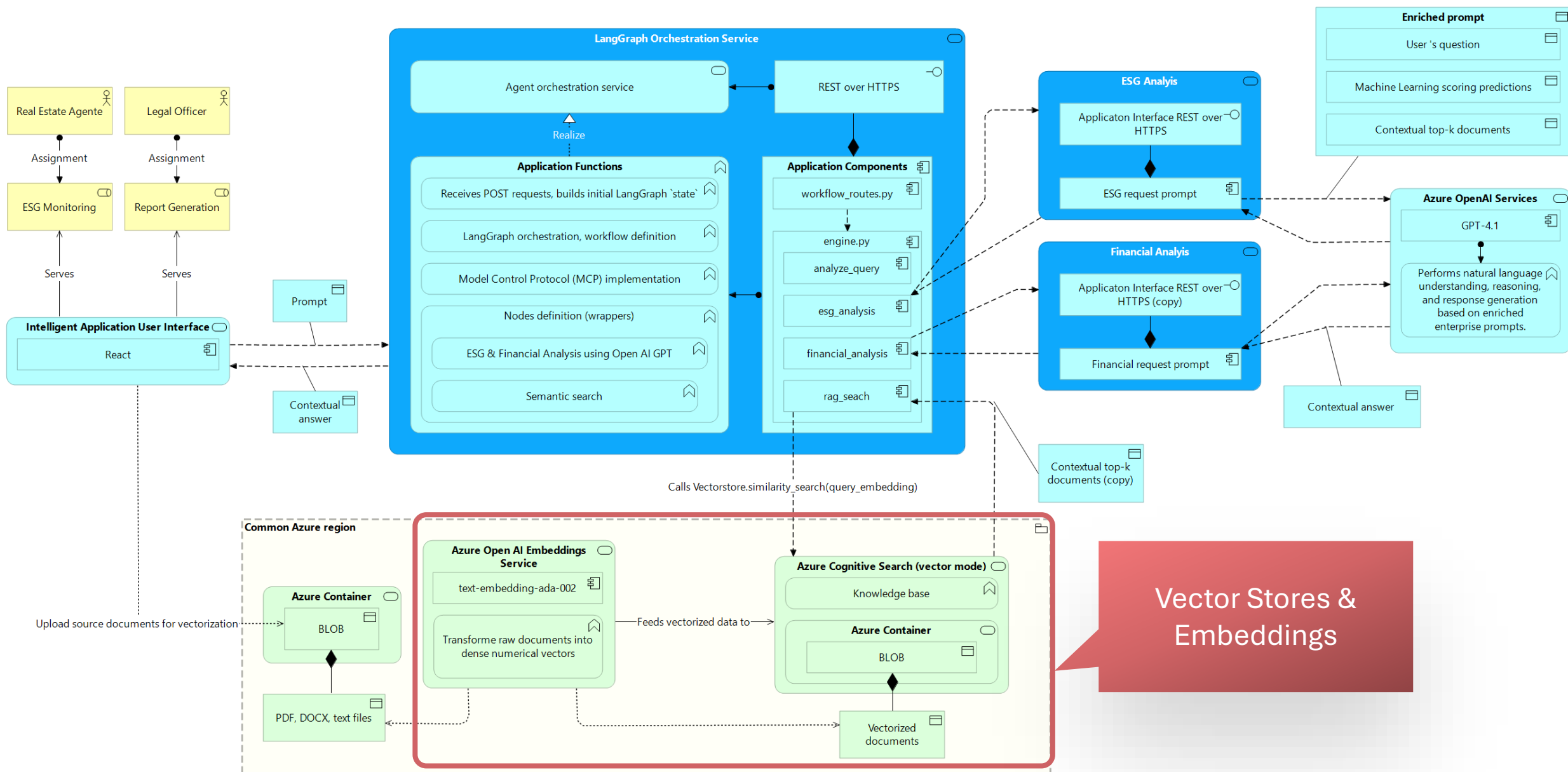
- LLMs are the “brain” behind tools like ChatGPT, capable of answering questions, generating content, translating text, summarizing information, and even reasoning
- Examples of LLM: OpenAI GPT-4.5, Google Gemini, Anthropic Claude, Meta LLaMa

2. Understand how vector stores and embeddings power semantic retrieval

*LLMs search by meaning, not keywords:
the foundation for RAG*



End-to-End Architecture of a Context-Aware AI Agent on Azure



Vector Stores & Embeddings

How LLMs Use Vector Stores to Find Meaningful Answers (the First Step in RAG)

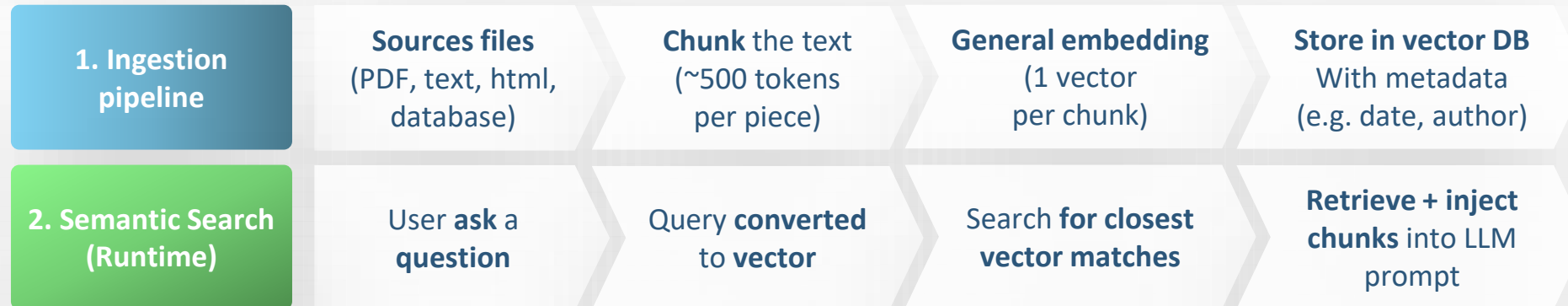
What is a Vector Store

- A **special type of database** that stores text as **semantic vectors** (embeddings)
- Enables search based on **meaning**, not just keywords

What are Semantic Vectors

- **Numerical representations** that capture the **meaning** of text
- **Similar meanings** $\Rightarrow$ **similar vectors** $\Rightarrow$ better retrieval

From documents to answers :



Retrieval quality depends on

- Chunk size: not too small, not too large
- Embedding model: quality impacts relevance
- Documents cleanliness: messy inputs = poor retrieval

A token is a unit of text used by language models

It can be a full word, part of a word, punctuation, space, or symbol

- Word $\rightarrow$ house
- Part of word $\rightarrow$ un, believ, able
- Punctuation $\rightarrow$., ?
- Space $\rightarrow$ ' '
- Symbol $\rightarrow$ #, 💡

Common Vector Store Technologies

Technology	Description / Use Case
◆ Azure Cognitive Search	• Cloud-native, enterprise-ready vector search with integrated security, filters, and metadata support • Ideal for RAG on Azure
◆ FAISS (Facebook AI Similarity Search)	• Fast and efficient local vector search library. • Optimized for large-scale similarity search • Often used in Python-based RAG systems.
◆ Pinecone	• Managed vector database with high scalability • Supports filtering, metadata, hybrid search. Cloud-hosted
◆ Weaviate	• Open-source or hosted vector DB with hybrid search (vector + keyword) • Offers GraphQL & REST APIs
◆ Qdrant	• Open-source vector database with strong support for filtering and similarity scoring
◆ Milvus	• High-performance open-source vector DB for massive datasets • Often used with GPU acceleration
◆ Redis (with vector support)	• Lightweight, real-time search (via Redis Search module), convenient when you already use Redis

Key Takeaways

Choose your vector store based on your context:

- **Azure Cognitive Search** for **enterprise-grade Azure solutions**
- **FAISS** / **Qdrant** for local, **open-source development**
- **Pinecone** / **Weaviate** / **Milvus** for scalable, **production-ready search**
- **Redis** for fast, real-time, **low-complexity use cases**

Model Deployment on Azure AI Foundry

Azure AI Foundry | Azure OpenAI / openai-ml-projects / Models + endpoints

Model deployments

Model deployments Service endpoints

+ Deploy model Refresh Edit Delete Open in playground Reset view

Name	Model name	Model version	State	Model retirement date	Content filter	Deployment type	Fine-tune	Capacity
gpt-4.1	gpt-4.1	2025-04-14	Succeeded	Apr 11, 2026 2:00 AM	DefaultV2 ⓘ	Global Standard		50
text-embedding-ada-002	text-embedding-ada-002	2	Succeeded	Apr 30, 2026 2:00 AM	DefaultV2 ⓘ	Standard		1

This shows the deployment of two essential models used in the RAG pipeline:

- [gpt-4.1](#): main LLM used for **reasoning, answering, and generating enriched responses**
- [text-embedding-ada-002](#): embedding model used to **convert documents into numerical vectors for similarity search**

Document Ingestion & Vectorization (RAG)

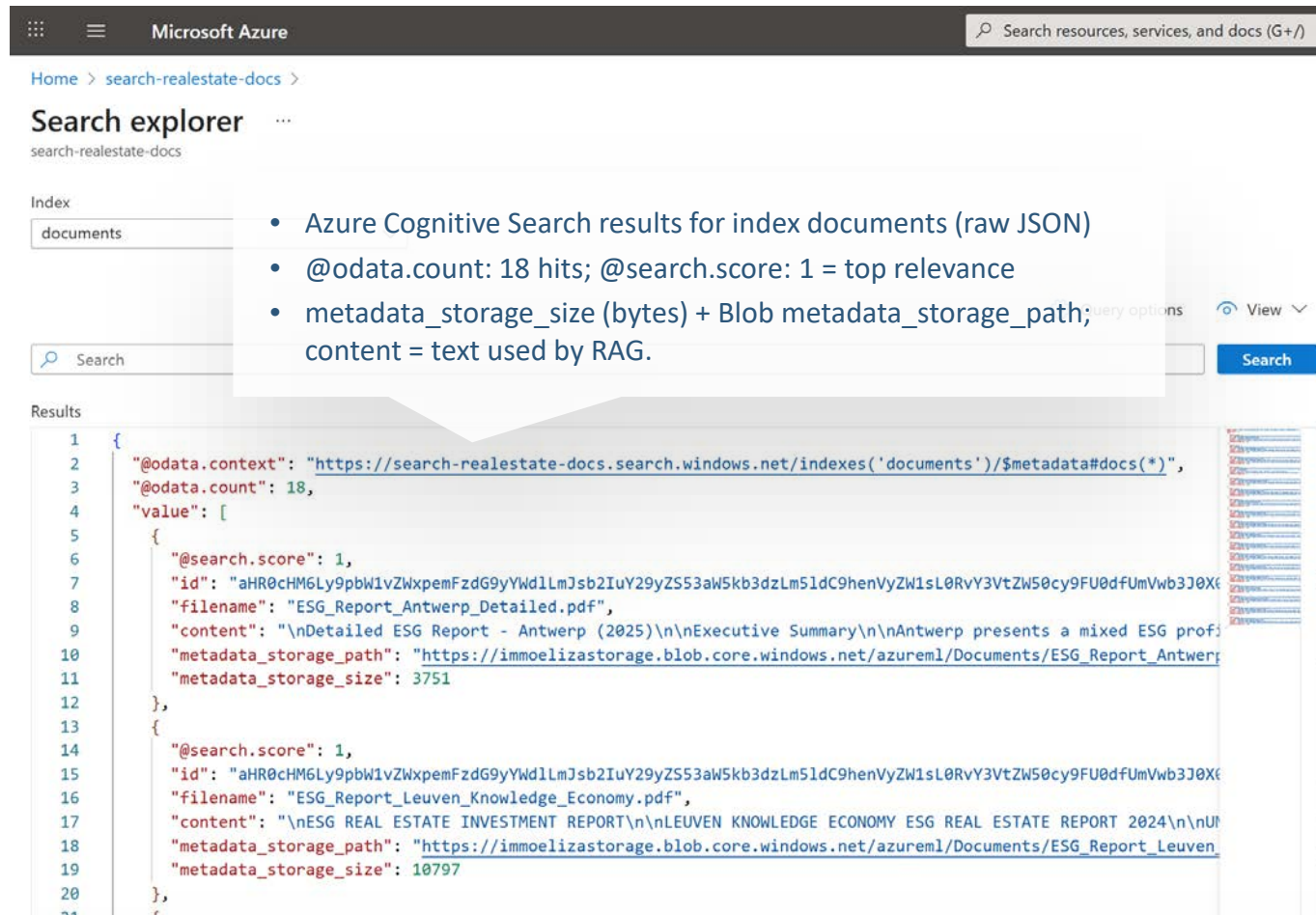
Upload ESG PDFs to Azure Blob, generate embeddings, and index in Azure Cognitive Search / FAISS

The screenshot shows the 'RAG Document Processing System' web portal. At the top, a green navigation bar contains links for 'RealEstate AI', 'Home', 'Price Predictor', 'ESG Agent', 'Documents RAG' (highlighted), 'Model Training', 'Admin Panel', 'Investment Manager', and 'Logout'. Below the navigation bar, the main heading is 'RAG Document Processing System' with the subtitle 'Upload your ESG, legal and energy documents for intelligent analysis'. A green status bar indicates 'Backend API connected (http://127.0.0.1:8010)'. The 'Vector Database' section features a toggle between 'Azure Cognitive Search' and 'FAISS'. The 'Upload & Vectorization' section shows 'Uploaded Documents (18)' with a legend for 'Uploaded to Azure Blob Storage' (blue square) and 'Vectorized & Indexed' (green square). Three documents are listed: 'ESG_Report_Antwerp_Detailed.pdf' (3.05 KB, PDF, Invalid Date), 'ESG_Report_Leuven_Knowledge_Economy.pdf' (9.02 KB, PDF, Invalid Date), and 'ESG_Report_Mons_Energy_Transition.pdf' (9.03 KB, PDF, Invalid Date). A callout box at the bottom states: 'The portal triggers upload → embedding → indexing with status badges (Uploaded, Vectorized & Indexed)'.

The screenshot shows the Microsoft Azure Storage browser interface for the 'immoelizastorage' storage account. The left sidebar lists various services, with 'Storage browser' selected. The main area displays the 'immoelizastorage' Blob containers. A list of 17 items is shown, including folders and PDF files. The PDF files are all named 'ESG_Report_...' followed by a location and 'Detailed.pdf' or 'Energy_Transition.pdf'. A callout box at the bottom states: 'The Blob container shows the source PDFs that feed semantic search'.

Powering RAG with Azure Cognitive Search

How LLMs Remember and Retrieve Knowledge



The screenshot shows the Microsoft Azure Search Explorer interface. The 'Index' dropdown is set to 'documents'. The search bar contains the text 'Search'. The results are displayed in a JSON format. A callout box highlights the following information:

- Azure Cognitive Search results for index documents (raw JSON)
- @odata.count: 18 hits; @search.score: 1 = top relevance
- metadata_storage_size (bytes) + Blob metadata_storage_path; content = text used by RAG.

The results show two documents:

```
1 {
2   "@odata.context": "https://search-realestate-docs.search.windows.net/indexes('documents')/$metadata#docs(*)",
3   "@odata.count": 18,
4   "value": [
5     {
6       "@search.score": 1,
7       "id": "aHR0cHM6Ly9pbW1vZWxpemFzdG9yYWdlLmJsb2IuY29yZS53aW5kb3dzLm5ldC9henVyZW1sL0RvY3VtZW50cy9FU0dfUmVwb3J0XG...",
8       "filename": "ESG_Report_Antwerp_Detailed.pdf",
9       "content": "\nDetailed ESG Report - Antwerp (2025)\n\nExecutive Summary\n\nAntwerp presents a mixed ESG profi...",
10      "metadata_storage_path": "https://immoelizastorage.blob.core.windows.net/azureml/Documents/ESG_Report_Antwerp...",
11      "metadata_storage_size": 3751
12    },
13    {
14      "@search.score": 1,
15      "id": "aHR0cHM6Ly9pbW1vZWxpemFzdG9yYWdlLmJsb2IuY29yZS53aW5kb3dzLm5ldC9henVyZW1sL0RvY3VtZW50cy9FU0dfUmVwb3J0XG...",
16      "filename": "ESG_Report_Leuven_Knowledge_Economy.pdf",
17      "content": "\nESG REAL ESTATE INVESTMENT REPORT\n\nLEUVEN KNOWLEDGE ECONOMY ESG REAL ESTATE REPORT 2024\n\nU...",
18      "metadata_storage_path": "https://immoelizastorage.blob.core.windows.net/azureml/Documents/ESG_Report_Leuven...",
19      "metadata_storage_size": 10797
20    }
21  ]
22 }
```

- The documents have been **ingested** and **chunked** for the **RAG pipeline**
- The **indexed content** is stored in Azure Cognitive Search and can be queried via
 - **Open Ai + LangChain**
 - Or **Azure SDK** (programmatically)
- **Semantic searches** allow to retrieve **high-relevance chunks** with citations
- This **enables secure, enterprise-grade memory** for **GenAI use cases** (audit, ESG, compliance, etc.)

- **OpenAI + LangChain** are used to build intelligent agents combining LLMs with **vector search**.
- **Azure SDK** allows full programmatic control of indexing and querying without LLM orchestration if needed

The screenshot shows the OpenAI Platform interface. On the left, a sidebar lists navigation options: Create (Chat, Audio, Images, Assistants), Manage (Usage, API keys, Logs, Storage, Batches), and Optimize (Evaluations, Fine-tuning). The main content area is titled 'API keys' and includes a 'Create new secret key' button. It states that the user has permission to view and manage all API keys in this project and provides a warning about not sharing API keys. A table lists the API keys:

NAME	SECRET KEY	CREATED	LAST USED	CREATED BY	PERMISSIONS
Secret key	sk-...qiga	Aug 13, 2025	Never	GoForContact	All

Below the table, a file explorer shows a folder named 'uploaded_documents' containing various PDF files, including ESG reports from Aalst, Antwerp, Arlon, Bruges, Brussels, Charleroi, Genk, Ghent, Hasselt, Leuven, Liege, Mechelen, Mons, Namur, Ostend, Tournai, and Verviers. An arrow points from this folder to a list of bullet points:

- The **OpenAI Embeddings API** is used to encode docs & queries into vectors
- Vectors are **indexed in FAISS** for semantic search powering RAG
- The API key is stored server-side in a Key Vault

Another arrow points from the bullet points to a file explorer showing a folder named 'faiss_index' containing 'index.faiss' and 'metadata.pkl'. Text next to this explains the FAISS index structure:

index.faiss: stores only **vector data** + integer IDs (**no business text**)

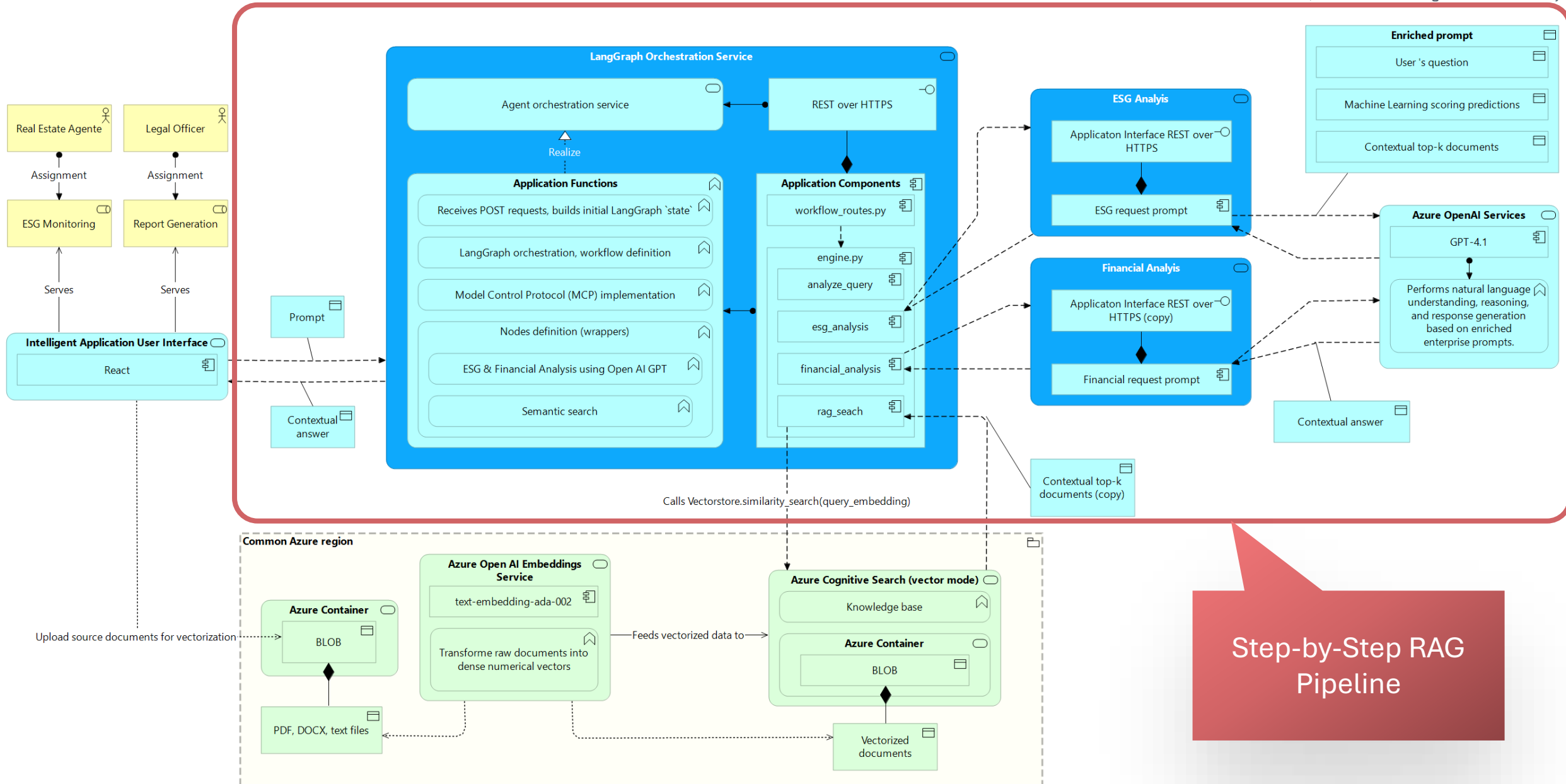
Metadata.pkl typical fields: doc_id, page, chunk_index, start_end_offsets, text_or_text_hash, embedding_model, dim, created_at

3. Follow the Retrieval-Augmented Generation (RAG) pipeline

From user intent to document search to grounded generation



End-to-End Architecture of a Context-Aware AI Agent on Azure



Introducing RAG pipeline (Retrieval-Augmented Generation)

- **RAG augments LLMs by retrieving relevant context** from **external data sources** (documents, wikis, DBs) at inference time
- It enables grounded, up-to-date, and domain-specific answers.

Step-by-Step RAG Pipeline

1. **User Intent (query or task)**
“Is this house in Antwerp compliant with Flanders’ EPC norms for 2030?”
2. **Embed Query**
Transformed into vector using `text-embedding-ada-002`
3. **Search the Vector Store**
Search performed on ESG compliance reports, EPC regulation PDFs, audit summaries (via Azure Cognitive Search)
4. **Retrieve Context (relevant document chunks) from Vector Store**
Regulatory excerpts: “By 2030, all homes must reach at least EPC label C...”
5. **Build the Prompt using Model Context Protocol (MCP)**
LLM receives “user intent + retrieved context + instructions”
6. **LLM generates the answer**
“This property (EPC = G) fails to meet 2030 standards and will require insulation and heating upgrades.”

Key Benefits

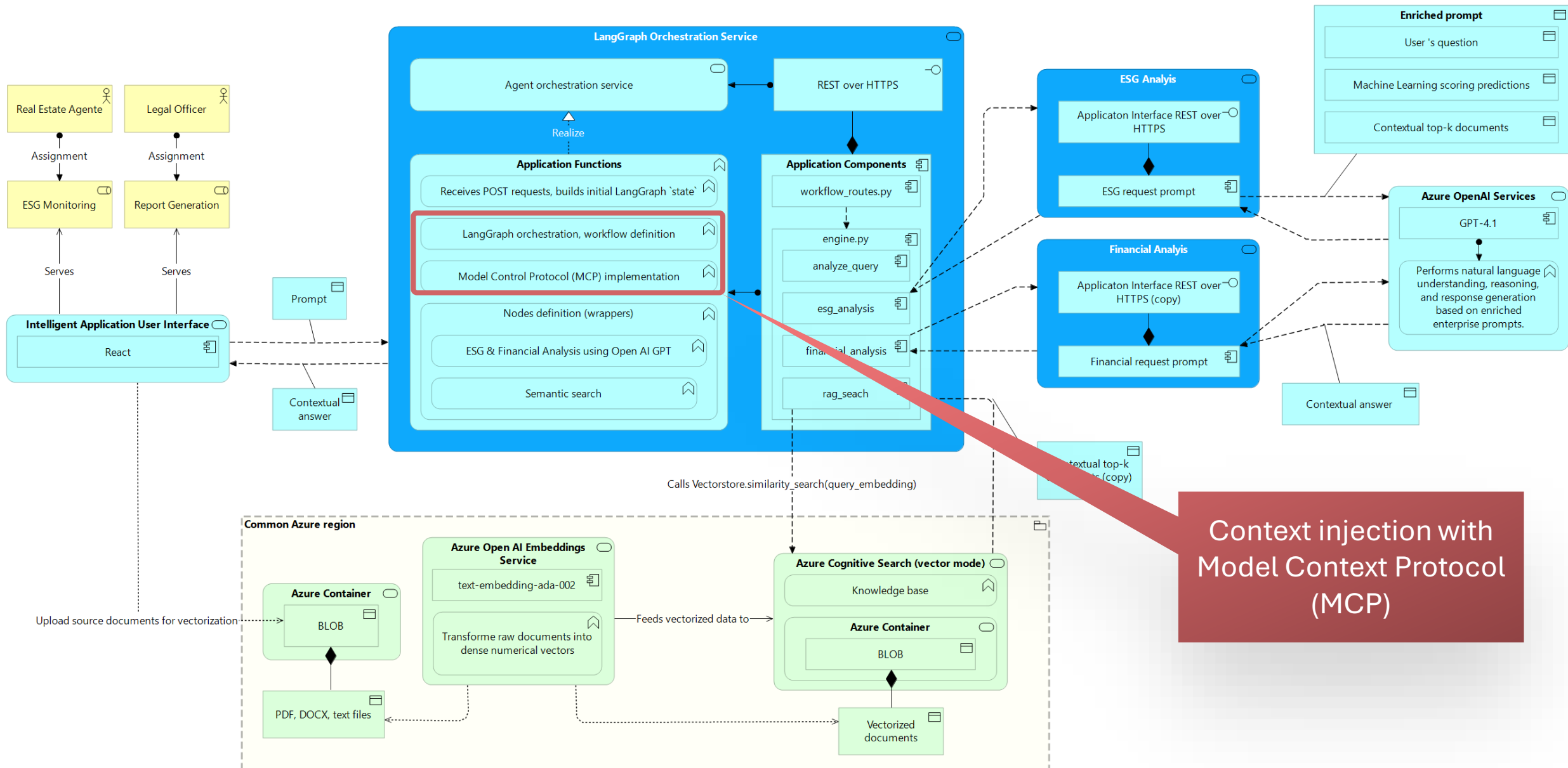
- **Reduces hallucinations** by grounding answers in authoritative sources
- **Keeps answers up-to-date** using live content — no need for costly model retraining
- **Connects LLMs to enterprise knowledge (PDFs, wikis, SharePoint, SQL, etc.)**
- **Enables explainability and control** by structuring the prompt explicitly via Model Context Protocol (MCP)
- **Cost-effective & scalable vs. fine-tuning**
 - ✓ No retraining required;
 - ✓ Scaling simply means uploading more docs
 - ✓ New users, new questions, new docs = no extra ML cost

4. How the Model Context Protocol (MCP) structures context for the LLM

Compare MCP implementations: LangChain, Semantic Kernel, or Custom Code



End-to-End Architecture of a Context-Aware AI Agent on Azure



Model Context Protocol (MCP) – Operationalizing Context for LLMs

What is MCP

- Defines how to select, structure, and **inject context** into a **prompt** for a Large Language Model (LLM)
- It answers: “**What** does the **model need to know**, and how should we tell it?”

Types of **Context** MCP Manages

- **System Instructions** : *"You are an ESG advisor..."*
- **User Profile & Roles** :
Analyst, Legal Officer, Facility Manager
- **Current Task** : *"Assess CO₂ compliance for Antwerp buildings"*
- **Retrieved Knowledge**:
Top-k vector DB chunks (via RAG)
- **Chat History**: *"As discussed earlier..."*
- **Document Metadata**:
Source, timestamp, regulation reference

Key benefits

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How is the Model Context Protocol (MCP) Implemented?

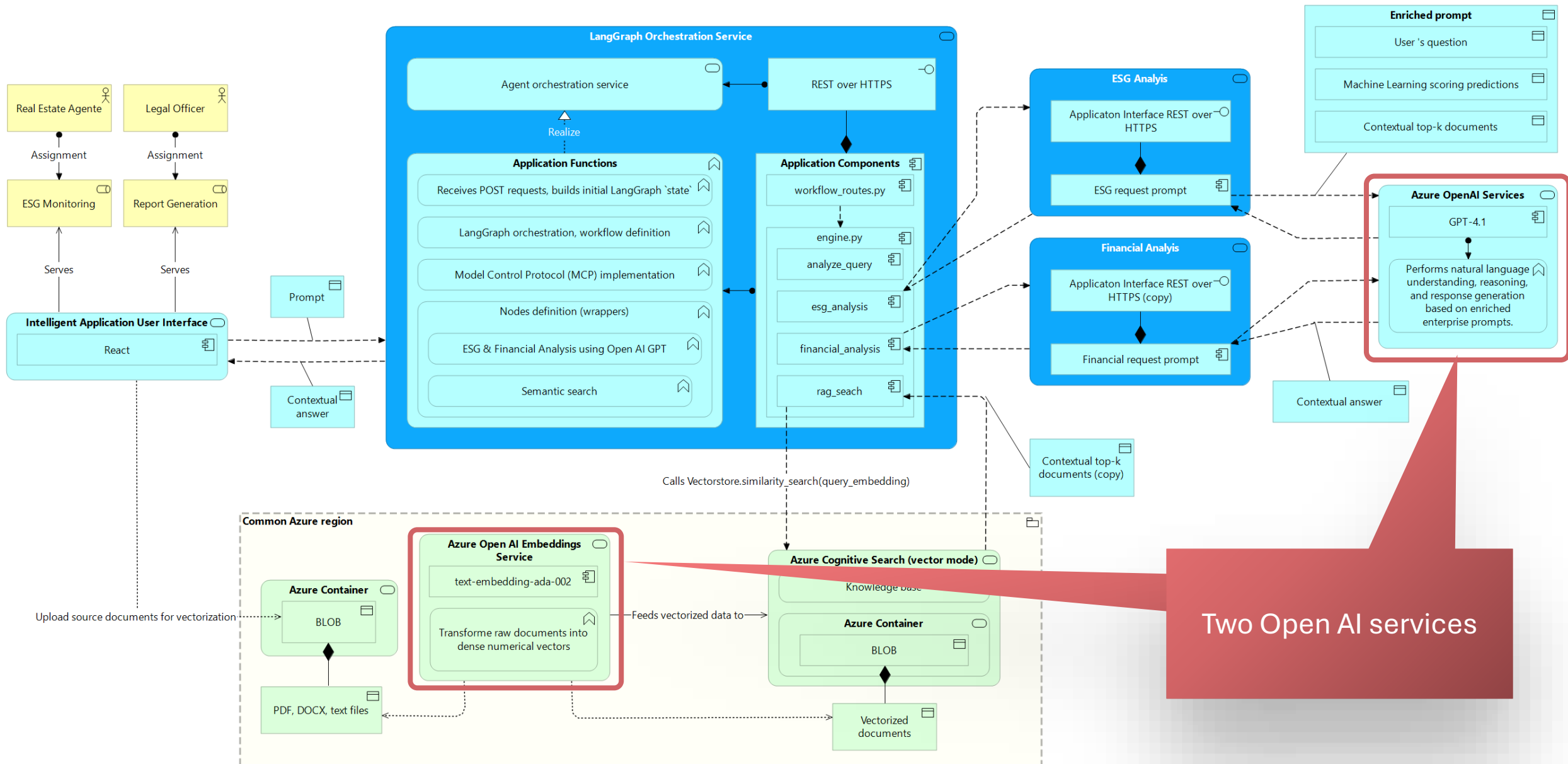
Approaches to Operationalize Context Injection in LLM Pipelines

Approach	Description & Capabilities	Best For
 LangChain	- Uses PromptTemplate, Memory, and RunnableSequence to build dynamic chains combining retrieval, user context, and tools. MCP is only partially supported, limited to local memory per chain.	- Composable, agentic workflows (RAG + Tools + Agents) - Suitable for simple, linear agent workflows.
 LangGraph	Orchestrates stateful, multi-step workflows with context-aware agents. MCP is natively integrated, enabling shared memory and context across agents and graph nodes.	Complex, multi-agent orchestration with branching logic , retries , and persistent context sharing .
 Semantic Kernel	- Orchestrates modular skills using Prompt Templates, a Planner, and Memory. - Context injection is handled through reusable planning and memory patterns.	Task-based LLM applications in Microsoft 365 (<i>Microsoft Graph, Outlook, Teams, SharePoint, Planner, OneDrive</i>) - and Azure (<i>Azure OpenAI, Cognitive Search, Blob Storage</i>) contexts
 Custom Python	- Full control over prompt injection and logic. - Developers manage formatting, metadata, context selection, and chaining manually.	Lightweight or highly customized use cases requiring maximum developer control, but minimal structure

Key Takeaways

-  **LangChain** offers **prompt** and **memory chaining**, but **only supports local context** without shared MCP across agents
-  **LangGraph** enables **native graph-based MCP orchestration** with **shared state** for complex, **multi-agent workflows**
-  **Semantic Kernel** supports **MCP** via **planners**, a sequence of **skills** (calendar, search), well-suited for **Microsoft 365** and **Azure-native scenarios**
-  Custom Python offers full flexibility to implement MCP manually, but requires more effort and governance

End-to-End Architecture of a Context-Aware AI Agent on Azure



Model Deployment on Azure AI Foundry

Azure AI Foundry | Azure OpenAI / openai-ml-projects / Models + endpoints

Model deployments

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Open AI SDK – The Gateway to GPT Reasoning & Embedding

What is the Open AI SDK

A software development kit (SDK) that **enables developers to interact with OpenAI's models (GPT-4, embeddings, function calling)** through simple, programmatic interfaces.

What Can You Do With Open AI SDK?

- **Call GPT-4.1 models**
 - Send prompts and receive answers, stream responses, or execute function calls.
- **Generate Embeddings**
 - Vectorize text using models like `text-embedding-ada-002`.
- **Manage Conversations & Tools**
 - Maintain message history, inject system prompts, define tools/functions for the LLM.
- **Streamline Integration**
 - Easily integrate with **LangChain, RAG agents, Azure Functions**, and more.

Typical Use Case

- **Inject prompts and receive streamed responses** from GPT-4.1
- **Generate embeddings** for semantic search and retrieval
- Call functions from GPT agents to trigger actions or retrieve data
- **Maintain conversation state** across multi-turn agent interactions
- **Integrate LLM outputs** into orchestrated **RAG pipelines**
- **Deploy agents** in cloud-native apps using Azure Functions or **APIs**

Feature / Capability	LangGraph (LangChain ecosystem)	OpenAI SDK
Purpose	High-level orchestration framework for agentic workflows	Low-level SDK to interact directly with OpenAI API endpoints
Prompt Engineering	[YES] PromptTemplates, Chains, Memory, MCP support	[INFO] Manual prompt construction with full flexibility
Context Injection (MCP)	[YES] Via shared state/checkpointed graph	[INFO] Developer must handle context formatting manually
Document Retrieval / RAG	[YES] Integrated with VectorStores, Retrievers, and Tools	[INFO] Needs external code for retrieval, injection, and chaining
Agents & Tools	[YES] Built-in agent abstractions (first-class multi-agent graphs)	[NO] No agent abstraction — must build orchestration manually
Function Calling	[YES] Integrated via tool wrapping or OpenAI tool-calling support	[YES] Native function calling supported (OpenAI 2023+ models)
Streaming Support	[YES] Full support with async components	[YES] Native support (stream=True)
Azure Integration	[YES] AzureOpenAI wrapper supported	[YES] Full support with azure_openai package and endpoint config
Use Case Fit	Complex, stateful, multi-agent RAG workflows	Lightweight, ideal for custom scripts or low-latency APIs
Learning Curve	Medium to high	Low to medium

- Use **LangGraph** when **building complex, multi-turn agents, orchestrating RAG pipelines, or injecting rich context using MCP**
- Use **OpenAI SDK** when you need a **lightweight interface for fast API calls**, quick prototypes, or want full control over the prompt structure.

Criteria	LangGraph (LangChain ecosystem)	Semantic Kernel (SK)
Origin / Ecosystem	Open source, agnostic (multi-cloud, OpenAI, Anthropic, etc.)	Microsoft open-source, optimized for Azure + Microsoft 365
Orchestration Logic	Manual chaining with RunnableSequence, memory, tools, function calling	Task planning via skills and automatic planner orchestration
Prompt Handling	PromptTemplate, multi-input/output chains	Semantic functions with YAML templates + structured parameters
Memory	Long-term memory (JSON, Redis, vector DB), short-term memory via buffer	Integrated memory via SK abstractions (embedding, retrieval, indexing)
Task Decomposition	Hand-coded multi-step chains (or agents)	Dynamic task planning with planner component
Integration Strength	Strong ecosystem: OpenAI, HuggingFace, LangSmith, vector stores	Native connectors to Microsoft Graph , SharePoint, Outlook, Teams, Azure AD
Best Use Cases	Complex agent workflows , tool chaining , advanced RAG pipelines	Enterprise assistants , copilots, knowledge workers using Microsoft tools
Development Language	Python (very active), JS/TS (less mature)	C# (mature), Python (evolving rapidly)
Flexibility	High: everything is composable, reusable	Medium: guided orchestration via modular skills
Learning Curve	Moderate to steep (requires chaining logic mastery)	Easier start with skills, YAML prompts, code fallback

- Use **LangGraph** when you need fine-grained control, **multi-agent logic**, and open ecosystem flexibility
- Use **Semantic Kernel** when you're in a **Microsoft 365 / Azure stack** and want modular skills, automatic planning, and fast integration with internal tools

5. Context-Aware Generation (CAG), combination of

- Knowledge retrieval (RAG)
- Task-aware prompt orchestration (MCP)
- Autonomous decision & action (Agents)



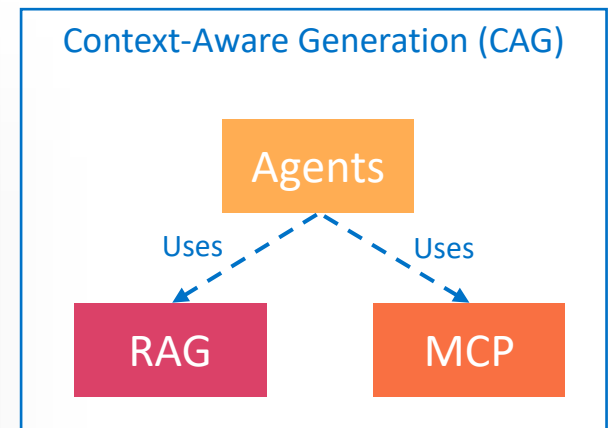
From Retrieval to Action: Structuring Context-Aware Generation (CAG)

Layer	Function	Technology	Core Role
 Retrieval-Augmented Generation (RAG)	Retrieves relevant knowledge chunks from enterprise data to enrich LLM prompts	Azure Cognitive Search, Embeddings	Ground the model's response with domain-specific context
 Model Context Protocol (MCP)	Structures and injects contextual information (user role, task, memory, metadata) into the prompt	LangChain, Semantic Kernel , Custom MCP	Enables dynamic, explainable, and structured prompt injection
 Agents	Reason, plan, and act autonomously by using MCP + RAG	AI Tuner Agent, ESG Agent, Autonomous Orchestrators	Perform goal-directed actions using tools, memory, and retrieved context

Key Takeaways

-  RAG = Passive retrieval  Enriches prompt with external knowledge
-  MCP = Active context orchestration  Injects structured, task-aware context
-  **Agents** = Autonomous execution  Reason and act **using both RAG + MCP**

Together, they enable **Context-Aware Generation (CAG)**: intelligent outputs aligned with business goals

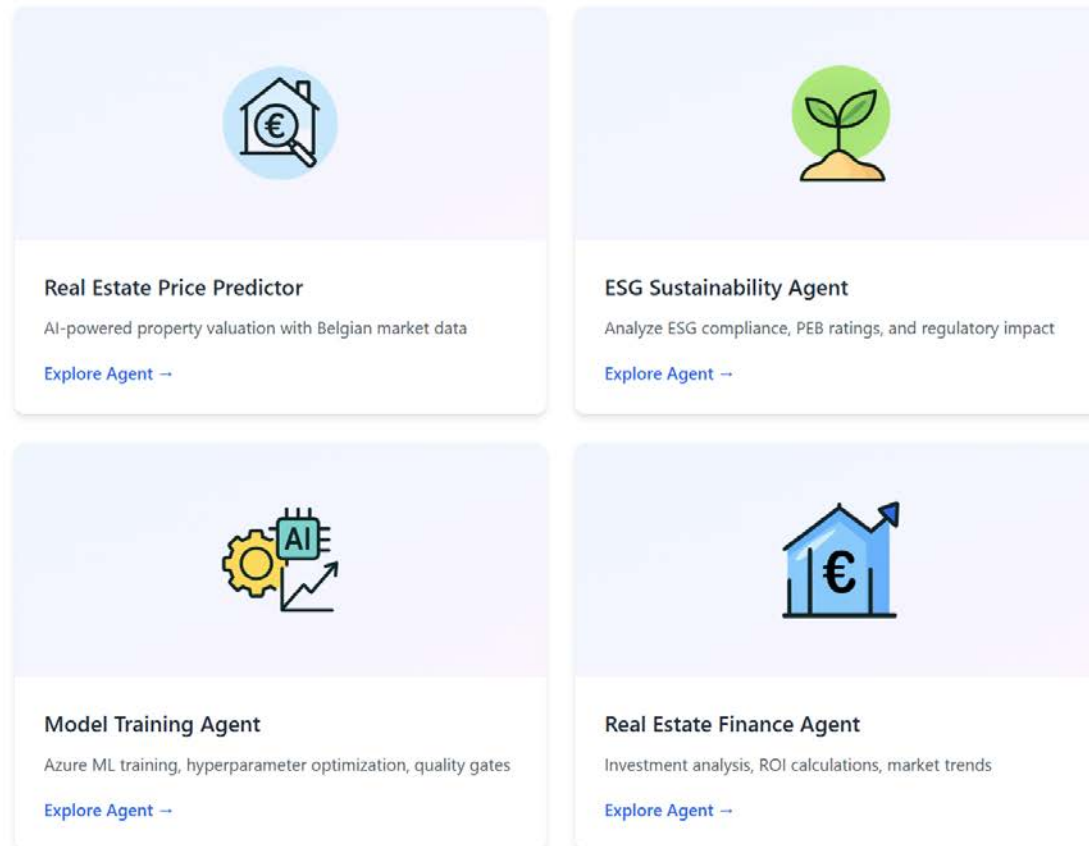


Multi-Agent Platform: Four Specialized Real-Estate Agents



Real Estate AI Platform

Comprehensive AI-powered real estate analysis platform with price prediction, ESG compliance, and advanced training capabilities.



Example of four agents :

- Price **Predictor**,
- **ESG** Sustainability,
- **Model** Training,
- and Finance agents

⇒ each handle a focused task

The agents

- Run on **shared context**
- **Orchestrated together** to deliver end-to-end analysis

⇒ Tuning	⇒ Valuation
⇒ Compliance	⇒ ROI

Profile: Yves

AI Chat Assistant

ESG Strategic Analysis

Clear

providing clear path and remediation options:

- MITIGATION** Fix interest rates where possible, stress-test cash flows at 1–2% higher rates, build liquidity buffer equivalent to six months' expenses. Operational Risks (Maintenance, Vacancy, Tenant Issues)
- "Risks:"** Deferred maintenance can compromise AS_NEW status; vacancy risk if energy standards not met.
- MITIGATION** Implement predictive maintenance software, offer short-term incentives to attract tenants, maintain responsive property management. Climate and Environmental Risks
- "Risks:"** Antwerp is prone to heatwaves and occasional flooding; electric heating may strain during peak demand, increasing costs and emissions.
- MITIGATION** Upgrade to energy-efficient heating systems, install passive cooling features (shading, ventilation), consider flood resilience measures for ground-floor areas. By systematically addressing these risks and capitalizing on market opportunities, the property can be repositioned as a resilient, ESG-compliant asset in the competitive Antwerp landscape.

Short-term Actions (0-6 months) - Electrical Heating Audit: Conduct a specialist review of the electric heating system. Identify quick wins such as installing smart thermostats, optimizing insulation around windows and doors, and replacing outdated radiators with more efficient models.

- Cosmetic Upgrades:** Minor enhancements (painting, LED lighting, high-efficiency water fixtures) will boost tenant appeal and rental value with minimal capex.
- Priority Maintenance:** Inspect and address any latent defects, particularly in the roof, windows, and external envelope, to prevent costly repairs and maintain AS_NEW condition.

Medium-term Strategy (6-24 months) - Energy Efficiency Upgrades: Based on EPC results, plan and implement insulation improvements (walls, roof, floors), consider replacing electric heating with hybrid or heat pump solutions, and install double-glazed windows. Secure available Flemish subsidies for energy renovations.

- Market Positioning Optimization:** Rebrand marketing to highlight improved energy performance and ESG upgrades. Target expatriate professionals and families, emphasizing proximity to schools, public transport, and green spaces.
- Smart Home Integration:** Install smart metering and home automation systems to further reduce energy consumption and enhance tenant experience.
- Governance Strengthening:** Formalize maintenance schedules and implement transparent reporting protocols to maintain high governance standards, positioning the property for institutional-grade investment.

Long-term Vision (2+ years)

- Regulatory Compliance Preparation:** Anticipate tightening of EPC requirements and potential carbon taxes; maintain a rolling compliance audit and budget for future upgrades.

Ask your question...

Send

ACTION TIMELINE

Strategic implementation roadmap

ST

SHORT TERM (1-3 MONTHS)

CRITICAL: Immediate compliance action, expert consultation, urban market analysis

URGENT

MT

MEDIUM TERM (3-12 MONTHS)

Emergency energy overhaul, regulatory compliance

MODERATE

LT

LONG TERM (1-5 YEARS)

Total energy renovation, property value recovery, premium market positioning

PLANNED

Executive Summary

STRATEGIC ANALYSIS

Strategic analysis for house in Antwerpen reveals a high-risk requiring immediate action investment opportunity. The property demonstrates critically poor energy performance and is significantly below market standards with potential restrictions. Key focus areas include emergency energy overhaul and full regulatory compliance. Established property requiring targeted modernization efforts.

Key Strategic Insights

✓ CRITICAL:

Immediate energy efficiency overhaul required

✓ Extremely high energy costs - major financial burden

✓ Property may face rental/sale restrictions without improvements

Analysis completed: 2025-08-11 22:00:34

Confidence Level: 0.85

ESG Analysis Report

HOUSE in Antwerpen, Antwerp

Detailed Analysis

36 insights generated

ESG Performance Scores

Environmental: 3.5/10

Social: 8.8/10

Governance: 8/10

Overall Score: 6.8/10

Current Status:

- EPC Rating:

G (very poor; indicates high energy consumption and low efficiency)

- Heating System:

Electric (often inefficient and expensive unless paired with renewable sources)

- Flood Risk:

Non-flood zone (positive for environmental resilience)

- Building Age/Condition:

Built in 2000, "as new" (structure likely sound, but energy systems outdated)

- Environmental Score:

3.5/10 (low)

Key Improvement Opportunities:

- Upgrade Insulation:

Install or improve wall, roof, and floor insulation to reduce heat loss.

- Window Replacement:

Replace any single glazing with high-efficiency double or triple glazing.

Disclaimer:

Generated by AI for informational purposes. Please consult with ESG experts for detailed assessments.

Context Demo – AI Property Report & ESG Insights

AI Property Report

Value, ESG & Compliance

ResetPrice PredictionESG Strategic Analysis

Predicted price : 620 592 €

Property Type

HOUSE

Subtype

HOUSE

Province

Antwerp

Locality

Antwerpen

Post Code

2000 – Antwerpen

Living Surface (M²)

180

Construction Year

2000

Bedrooms

3

Rooms Total

5

Bathrooms

1

Toilets

1

Facade Count

2

Building Condition

AS NEW

Kitchen Type

HYPER EQUIPPED

Heating Type

ELECTRIC

Flood Zone

NON FLOOD ZONE

EPC Score (Energy Class)

G

Additional Features

☒ Has Living Room ☒ Has Terrace

ESG Scores Summary

3.5/10

Environmental

8.8/10

Social

8/10

Governance

6.8/10

Overall ESG

ESG Score Calculation

Our ESG scores are calculated based on property characteristics and market data:

Environmental: Energy efficiency (EPC rating), heating systems, flood risk, and carbon footprint

Social: Location quality, accessibility, community services, and neighborhood safety

Governance: Legal compliance, building regulations, property management standards

Overall: Weighted average of all three pillars with emphasis on environmental factors

8-10: Excellent

6-8: Good

3-6: Fair

0-3: Poor

This form is the **starting point of the agent workflow**. It provides property-specific inputs, such as :

- surface area,
- number of rooms,
- EPC class,
- location
- and price prediction

ESG Analysis Report (Context-Aware Output)

- This report is generated by the ESG Agent, combining **form inputs** with **contextual knowledge** (e.g., EPC rules, energy costs, local data).
- It delivers **tailored insights** (here, for a G-rated house in Antwerp), using **retrieved documents** and **domain-specific rules**.

💡 *Added value:*

Not a generic answer, but a **grounded, personalized analysis** powered by RAG, helping users make **informed ESG decisions**.

ESG Analysis Report

HOUSE in Antwerpen, Antwerp

Detailed Analysis

8 insights generated

1. EPC Rating & Energy Performance

The property's EPC score of G represents the lowest energy label in Belgium, signifying very poor energy efficiency. In Flanders, and specifically in Antwerp, the average EPC for houses constructed post-2000 is typically between C and D, meaning this property underperforms even among its peers. Flanders has set ambitious energy performance targets, aiming for all residential buildings to reach at least EPC label D by 2030 and label A or B by 2050. Therefore, this property falls significantly short of both current expectations and future mandatory standards. Its G rating indicates substantial inefficiencies, likely due to inadequate insulation, outdated windows, or inefficient heating systems, despite its recent construction and "as new" condition.

2. Estimated Energy Consumption & Costs

Based on the EPC G rating, expected primary energy consumption for a house of this size (180 m²) is typically in the range of 500–600 kWh/m²/year, resulting in an annual total consumption of approximately 90,000–108,000 kWh. Given current Belgian residential electricity rates (averaging €0.35/kWh), total annual energy costs could range from €31,500 to €37,800, which is exceptionally high and well above the national average for comparable homes. These estimates assume continued use of the existing electric heating system and do not account for possible behavioral or technical optimizations. Such high operational costs would be unsustainable for most occupants and represent a significant financial risk.

Disclaimer: Generated by AI for informational purposes. Please consult with ESG experts for detailed assessments.

Context-Aware GenAI for the Enterprise: RAG, Agent Design & Modular Orchestration

Yves Schillings

31

Profile: Yves

AI Chat Assistant

ESG Strategic Analysis

Clear

on the electric heating system and insulation performance.

- Quick-win Upgrades: Install LED lighting, smart thermostats, and water-saving fixtures to improve energy and resource efficiency with minimal investment.
- Maintenance Priorities: Review and service all critical building systems (roof, windows, electrical wiring, ventilation) to ensure AS, NEW status is genuine and to prevent latent defects.
- Market Repositioning: Target marketing efforts towards young professionals and families seeking move-in-ready properties, emphasising condition and centrality, while transparently communicating ongoing ESG improvements.

Medium-term Strategy (6-24 months) - Heating System Upgrade: Replace or supplement conventional electric heating with high-efficiency heat pump technology, which will substantially improve EPC and lower operating costs.

- Building Envelope Enhancement: Invest in superior insulation (roof, walls, windows) to achieve at least EPC C or higher, future-proofing against tightening standards and enhancing tenant comfort.
- Smart Home Integration: Implement smart meters and energy management solutions to appeal to tech-savvy tenants and enable more efficient resource use.
- Sustainable Amenities: Consider on-site renewable energy (e.g., photovoltaic panels) where feasible, and improve recycling/waste management infrastructure.
- Market Positioning Optimisation: Rebrand the property as a sustainable, low-carbon residence, enabling premium rental pricing and attracting ESG-oriented tenants.
- Stakeholder Engagement: Engage with local authorities regarding upcoming regulatory changes and potential subsidies for energy retrofits.

Long-term Vision (2+ years) - Future-Proofing: Monitor and prepare for the anticipated tightening of Flemish energy standards (e.g., EPC B or higher by 2030), including staged investment in further efficiency measures.

- Regulatory Compliance: Develop a compliance roadmap, ensuring the property stays ahead of evolving rental and sales eligibility criteria tied to EPCs and carbon performance.
- Portfolio Expansion: Leverage improved ESG credentials and performance data to access green financing for further acquisitions in Antwerp or other high-growth Belgian urban centres.
- Tenant Retention & Community Integration: Foster a community-oriented living environment through regular tenant engagement, wellness amenities, and partnerships with local service providers, strengthening the property's social and governance dimensions.
- Data-Driven Asset Management: Utilize digital monitoring of building performance to inform ongoing maintenance and capital planning, maximising value and minimising operational surprises.

Ask your question...

Send

ESG Agent Output (Strategic Dialogue)

- The AI Agent **generates a detailed strategy** (short, mid, long-term), combining user input with contextual data (building specs, EPC rules, local policies).

💡 Added value:

Translates ESG constraints into **clear strategic recommendations** tailored to the property and timeline.

Executive Summary & Roadmap

- This is the summarized, **decision-ready output**, **grounded in prior analysis**.
- It highlights **urgency, risks**, and **next steps** (timeline, strategic impact).

💡 Added value:

Enables **prioritization** and **decision-making** by condensing **complex ESG factors** into **clear guidance for action**.

ACTION TIMELINE

Strategic implementation roadmap

ST

SHORT TERM (1-3 MONTHS)

CRITICAL: Immediate compliance action, expert consultation, urban market analysis

URGENT

MT

MEDIUM TERM (3-12 MONTHS)

Emergency energy overhaul, regulatory compliance

MODERATE

LT

LONG TERM (1-5 YEARS)

Total energy renovation, property value recovery, premium market positioning

PLANNED

Executive Summary

STRATEGIC ANALYSIS

Strategic analysis for house in Antwerpen reveals a **high-risk requiring immediate action** investment opportunity. The property demonstrates **critically poor energy performance** and is **significantly below market standards with potential restrictions**. Key focus areas include **emergency energy overhaul and full regulatory compliance**. Established property requiring **targeted modernization efforts**.

Key Strategic Insights

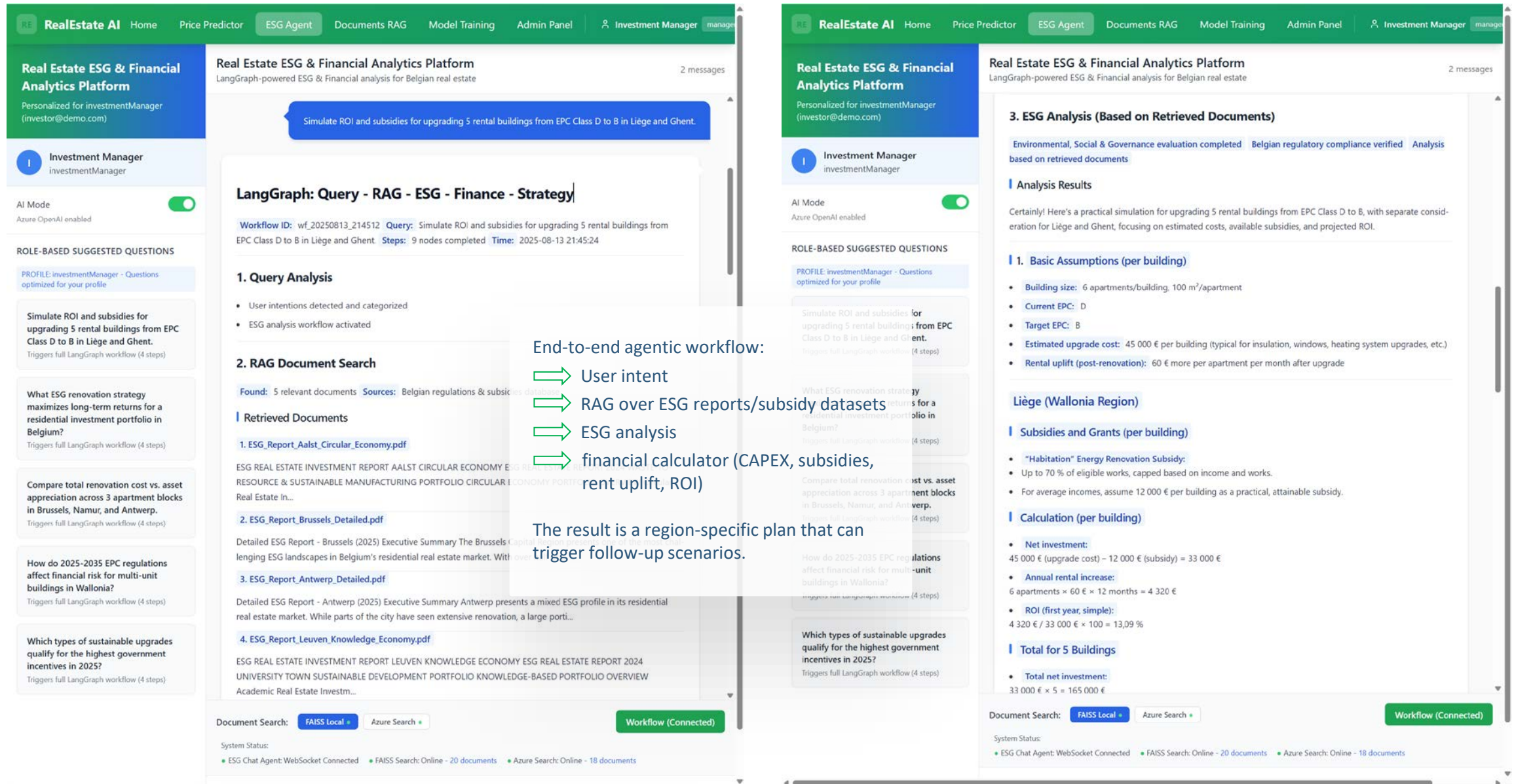
- ✓ **CRITICAL:** Immediate energy efficiency overhaul required
- ✓ **Extremely high energy costs** - major financial burden
- ✓ Property may face **rental/sale restrictions** without improvements

Analysis completed: 2025-08-09 21:42:47

Confidence Level: 0.85

Real Estate ESG & Financial Analytics Platform (1/2)

LangGraph: Query - RAG - ESG - Finance - Strategy



Real Estate ESG & Financial Analytics Platform
LangGraph-powered ESG & Financial analysis for Belgian real estate

2 messages

Simulate ROI and subsidies for upgrading 5 rental buildings from EPC Class D to B in Liège and Ghent.

LangGraph: Query - RAG - ESG - Finance - Strategy

Workflow ID: wf_20250813_214512 Query: Simulate ROI and subsidies for upgrading 5 rental buildings from EPC Class D to B in Liège and Ghent Steps: 9 nodes completed Time: 2025-08-13 21:45:24

1. Query Analysis

- User intentions detected and categorized
- ESG analysis workflow activated

2. RAG Document Search

Found: 5 relevant documents Sources: Belgian regulations & subsidies database

Retrieved Documents

- ESG_Report_Aalst_Circular_Economy.pdf
- ESG_Report_Brussels_Detailed.pdf
- ESG_Report_Antwerp_Detailed.pdf
- ESG_Report_Leuven_Knowledge_Economy.pdf

Document Search: FAISS Local Azure Search Workflow (Connected)

System Status:
• ESG Chat Agent: WebSocket Connected • FAISS Search: Online - 20 documents • Azure Search: Online - 18 documents

3. ESG Analysis (Based on Retrieved Documents)

Environmental, Social & Governance evaluation completed Belgian regulatory compliance verified Analysis based on retrieved documents

Analysis Results

Certainly! Here's a practical simulation for upgrading 5 rental buildings from EPC Class D to B, with separate consideration for Liège and Ghent, focusing on estimated costs, available subsidies, and projected ROI.

1. Basic Assumptions (per building)

- Building size: 6 apartments/building, 100 m²/apartment
- Current EPC: D
- Target EPC: B
- Estimated upgrade cost: 45 000 € per building (typical for insulation, windows, heating system upgrades, etc.)
- Rental uplift (post-renovation): 60 € more per apartment per month after upgrade

Liège (Wallonia Region)

Subsidies and Grants (per building)

- "Habitation" Energy Renovation Subsidy:
 - Up to 70 % of eligible works, capped based on income and works.
 - For average incomes, assume 12 000 € per building as a practical, attainable subsidy.

Calculation (per building)

- Net investment:
45 000 € (upgrade cost) - 12 000 € (subsidy) = 33 000 €
- Annual rental increase:
6 apartments × 60 € × 12 months = 4 320 €
- ROI (first year, simple):
4 320 € / 33 000 € × 100 = 13,09 %

Total for 5 Buildings

- Total net investment:
33 000 € × 5 = 165 000 €

Document Search: FAISS Local Azure Search Workflow (Connected)

System Status:
• ESG Chat Agent: WebSocket Connected • FAISS Search: Online - 20 documents • Azure Search: Online - 18 documents

End-to-end agentic workflow:

- ➡ User intent
- ➡ RAG over ESG reports/subsidy datasets
- ➡ ESG analysis
- ➡ financial calculator (CAPEX, subsidies, rent uplift, ROI)

The result is a region-specific plan that can trigger follow-up scenarios.

Real Estate ESG & Financial Analytics Platform (2/2)

LangGraph: Query - RAG - ESG - Finance - Strategy

Real Estate ESG & Financial Analytics Platform
Personalized for investmentManager (investor@demo.com)

Investment Manager
investmentManager

AI Mode
Azure OpenAI enabled

ROLE-BASED SUGGESTED QUESTIONS
PROFILE: investmentManager - Questions optimized for your profile

Simulate ROI and subsidies for upgrading 5 rental buildings from EPC Class D to B in Liège and Ghent.
Triggers full LangGraph workflow (4 steps)

What ESG renovation strategy maximizes long-term returns for a residential investment portfolio in Belgium?
Triggers full LangGraph workflow (4 steps)

Compare total renovation cost vs. asset appreciation across 3 apartment blocks in Brussels, Namur, and Antwerp.
Triggers full LangGraph workflow (4 steps)

How do 2025-2035 EPC regulations affect financial risk for multi-unit buildings in Wallonia?
Triggers full LangGraph workflow (4 steps)

Which types of sustainable upgrades qualify for the highest government incentives in 2025?
Triggers full LangGraph workflow (4 steps)

Real Estate ESG & Financial Analytics Platform
LangGraph-powered ESG & Financial analysis for Belgian real estate

2 messages

Summary Table

City	Net Investment (per building)	Subsidy (per building)	Total Net Investment (\$ buildings)	Annual Rental Uplift (\$ buildings)	ROI (ap-prox)
Liège	33 000 €	12 000 €	165 000 €	21 600 €	13,09 %
Ghent	33 750 €	11 250 €	168 750 €	21 600 €	12,80 %

Additional Practical Advice

- EPC Class B increases property value and future-proofs your assets against stricter rental regulations (especially in Flanders, where EPC requirements are tightening).
- Subsidy applications require pre-approval and invoices.
- Combine upgrades (roof, wall insulation, windows, heating) to maximize both energy savings and subsidy eligibility.
- Monitor regional updates: Subsidy levels and eligibility.

4. Financial Analysis

Renovation costs calculated and subsidies identified

Estimated Renovation Costs

Item
Standard energy improvement

Total costs: €25,000

Available Grants and Subsidies

Program	Amount (EUR)
---------	--------------

Document Search: FAISS Local Azure Search

System Status:
ESG Chat Agent: WebSocket Connected FAISS Search: Online - 20 documents Azure Search: Online - 18 documents

Workflow (Connected)

Real Estate ESG & Financial Analytics Platform
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Real Estate ESG & Financial Analytics Platform
LangGraph-powered ESG & Financial analysis for Belgian real estate

2 messages

Regional subsidies (combined)
€8,000

Total subsidies: €15,500

ROI Analysis

Metric	Value
Net cost after subsidies	€9,500.0
Property value increase	€13,300.0
Estimated ROI	40.0%
Net profit	€3,800.0
Payback period	999.0 years
Annual energy savings	€0.0

5. Strategic Recommendations

Action Plan

Priority	Recommendation	Timeline
1	Optimal timing: Start procedures in 2025 to benefit from subsidies	Q1 2025
2	Phased approach: Prioritize insulation then heating system	Q2-Q3 2025
3	2030 compliance: Ensure EPC regulations compliance	Before 2030
4	Tax optimization: Combine regional subsidies + federal tax reduction	Ongoing

Analysis generated by LangGraph ESG-Finance Engine 9 analysis steps completed

Report generated on 2025-08-13 at 21:45:24

Document Search: FAISS Local Azure Search

System Status:
ESG Chat Agent: WebSocket Connected FAISS Search: Online - 20 documents Azure Search: Online - 18 documents

Workflow (Connected)

Ask about ESG + Financial analysis, ROI, renovation costs, grants...
Send

LangGraph ESG-Finance Integration - Belgian regulations and ESG compliance

Live Orchestration Example – LangGraph ESG Workflow

Real Estate ESG & Financial Analytics Platform
Personalized for investmentManager (investor@demo.com)

Investment Manager
investmentManager

AI Mode
Azure OpenAI enabled

ROLE-BASED SUGGESTED QUESTIONS

PROFILE: investmentManager - Questions optimized for your profile

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Triggers full LangGraph workflow (4 steps)

Real Estate ESG & Financial Analytics Platform
LangGraph-powered ESG & Financial analysis for Belgian real estate

LangGraph ESG Workflow Visualizer

LangGraph ESG Workflow CONNECTED

Test Workflow Refresh Clear Trace

Analyze Query
Process and understand the user query for ESG analysis

RAG Search
Search through property documents and regulations

ESG Analysis
Perform comprehensive ESG analysis using AI models

Calculate Financials
Calculate financial metrics and ROI based on ESG analysis

Generate Response
Create comprehensive ESG analysis report with financial insights

Workflow Information

Name: **esg-agent-integration** Version: **2.0.0** Nodes: **5**

Execution Trace (Real-time)

10:29:30 PM Analyze Query Analyze Query completed successfully

10:29:31 PM RAG Search RAG Search completed successfully

10:29:31 PM ESG Analysis Esg Analysis completed successfully

10:29:31 PM Calculate Financials Calculate Financials completed successfully

10:29:31 PM Generate Response Generate Esg Finance Response completed successfully

3	2030 compliance: Ensure EPC regulations compliance	Before 2030
4	Tax optimization: Combine regional subsidies + federal tax reduction	Ongoing

Analysis generated by LangGraph ESG-Finance Engine 9 analysis steps completed

Report generated on 2025-08-13 at 22:41:59

Document Search: FAISS Local Azure Search

Workflow (Connected)

System Status:
ESG Chat Agent: WebSocket Connected ESG Search: Online - 20 documents Azure Search: Online - 18 documents

Ask about ESG + Financial analysis, ROI, renovation costs, grants... Send

- **LangGraph** pipeline (5 nodes)
- **WebSocket** step updates with timestamps
- Node-level success/fail
- Versioned & **connected**: **workflow v2.0.0**
- Supports **targeted retries**
- Audit-ready: log prompts, metrics, outputs per step

```
Step update: generate_esg_finance_response - completed LangGraphVisualizer.jsx:223
WebSocket message: LangGraphVisualizer.jsx:128
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WebSocket message: LangGraphVisualizer.jsx:128
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Workflow completed LangGraphVisualizer.jsx:229
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Phase 2 – Multi-Agent Integration

Move from a single isolated agent to a collaborative ecosystem of agents sharing context

6. Learn how AI agents use LLMs to reason, plan, and act autonomously

Modular, production-grade, and Azure-integrated



What are AI Agents ?

Why Prompting to Autonomous Problem Solving

AI Agent

- Intelligent systems that **use LLMs as their reasoning core**
- Enhanced with **additional components** that allow them to reason, plan and act autonomously
- Components: **memory**, APIs, **databases**, **planer** / tasks Managers, **RAG** modules, **orchestrators**, **evaluators**

Reason

To understand complex problems and **break them down into manageable sub-goals**

AI Agents Components

Plan

To determine a structured **sequence of steps toward** solving those goals

Act Autonomously

By interacting with tools, APIs, or systems, adapting based on feedback, and executing steps **without human intervention**

Example: AI Tuner Agent for Model Optimization

Scenario

We want to automatically tune a CatBoost-Optuna model for real estate price prediction
⇒ to **optimize parameters** with **minimal human intervention**

Plan

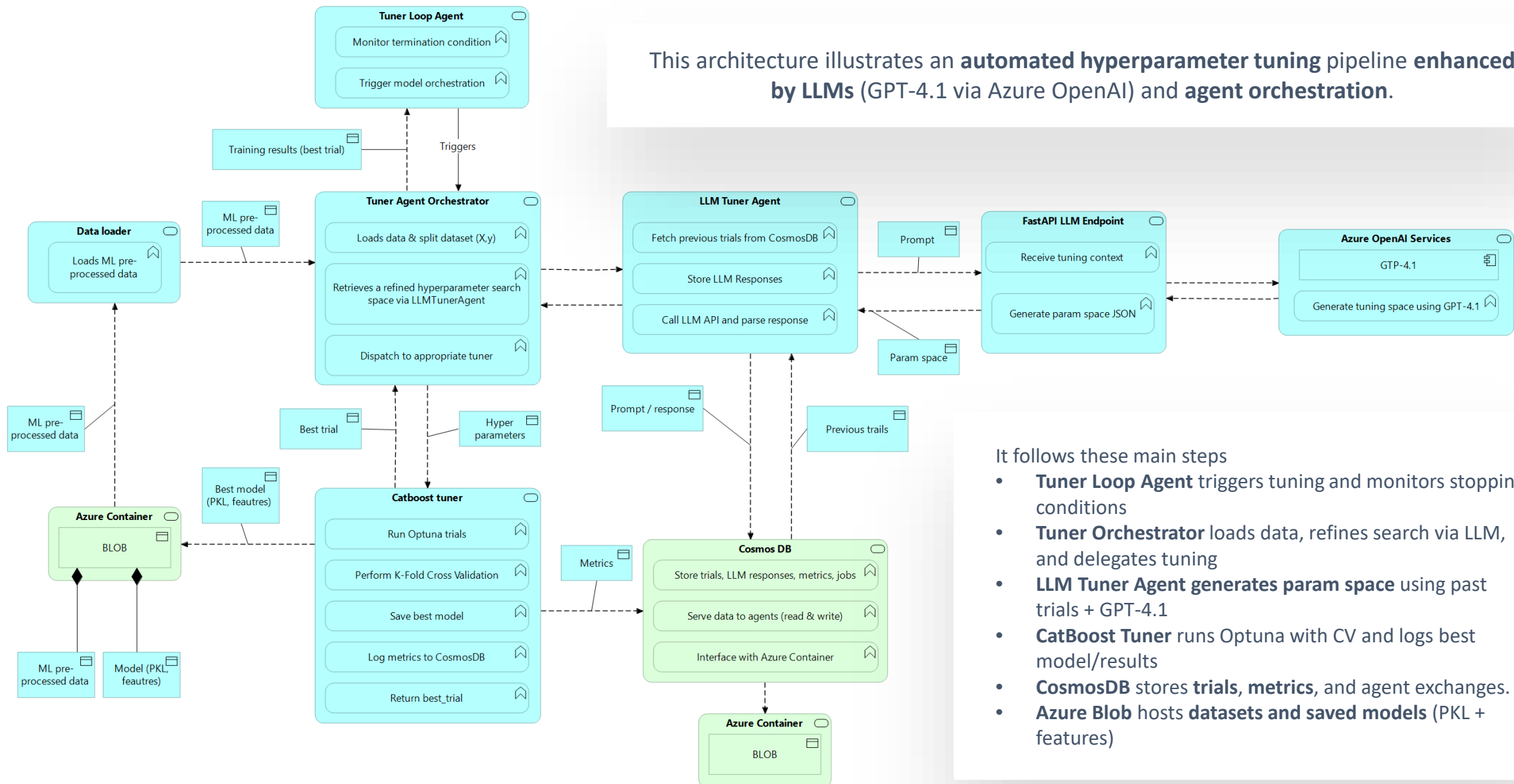
- **Load LLM-proposed parameter space**
- **Set guardrails:**
 - Early stopping
 - 5-fold cross-validation (Optuna)
 - Runtime/trial limits

Act Autonomously

- **Run tuning loop**, score each model, rank results
- **Persist best model** and **metrics** to Azure (Blob Storage + Cosmos DB)
- **Restart loop** with **updated context**

Reason

- **Past trials feed memory**
⇒ to refine the search each cycle
- Agents **reason, adapt, and execute without human help**
- **Continuous feedback** steadily **improves performance**



7. Multi-Agent Workflow

From single-agent context to shared, dynamic memory across orchestrated agents.



What is a Multi-Agent Workflow?

A coordinated system where multiple AI agents, each with a specialized role, **work together** through a **shared context** and **orchestrator** to complete complex tasks more efficiently than a single agent.

Key characteristics

- **Specialization:** each agent focuses on one domain (e.g., tuning, evaluation, compliance)
- **Shared Memory:** all agents have access to relevant context, history, and outputs from others
- **Autonomous Collaboration:** agents trigger and pass tasks to each other without manual intervention
- **Dynamic Adaptation:** workflows evolve as performance metrics and context change

Why it matters

- Handles complex, **multi-step processes** at scale
- **Reduces** human hand-offs and **duplicate work**
- **Enables** faster, more **accurate**, and **traceable decision-making**

Scaling LangChain Model Context Protocol (MCP) for Multi-Agent Model Tuning

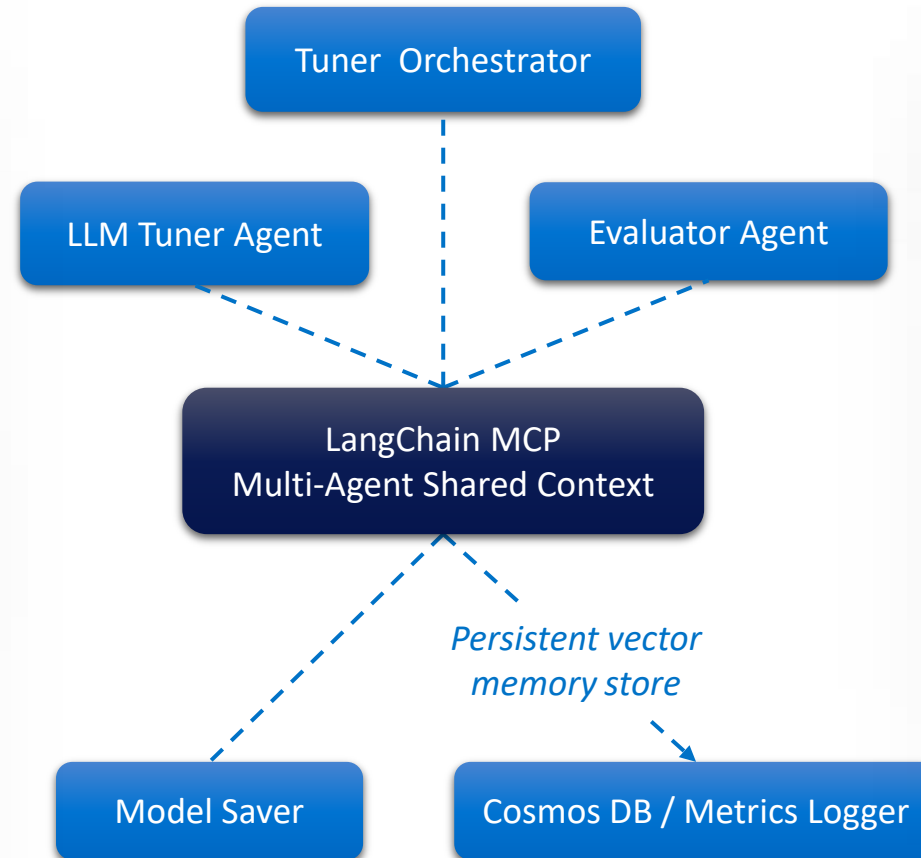
From single-agent context to shared, dynamic memory across orchestrated agents.

MCP Role

- Unified context for all agents (past trials, params, metrics)
- Role-specific filtered views per agent
- Continuous updates from metrics and evaluator feedback
- Prevents duplicate work and improves trial efficiency

Implementation

- Built using LangChain's MCP framework for shared memory and tool orchestration
- Acts as a dynamic message bus between agents
- Integrates with CosmosDB for persistent memory



Key Benefits

- Scalable: plug new agents easily (Feature Selector, Data Cleaner...)
- Traceable: full history of parameters & results
- Adaptive: context evolves with performance
- LangChain integration ensures modularity and rapid agent extension

Model Training & metrics

Home > neuromesh-cosmosdb

 **neuromesh-cosmosdb | Data Explorer** ☆ ...
 Azure Cosmos DB account

Enable Azure Synapse Link
Visual Studio Code
New SQL Query
Open Query
New Stored Procedure

Overview
 Activity log
 Access control (IAM)
 Tags
 Diagnose and solve problems
 Quick start
Data Explorer
 Mirroring in Fabric (Preview)
 Resource visualizer
 Settings
 Integrations
 Containers
 Browse
 Settings
 Monitoring
 Automation
 Help

+ New...

Home

cosmicworks
 products
 ModelTrainingDB
 ModelMetrics
 Items
 Settings
 Stored Procedures
 User Defined Functions
 Triggers
 TrainingJobs
 Items
 Settings
 Stored Procedures
 User Defined Functions
 Triggers
 TrainingLogs
 Items
 Settings
 Stored Procedures
 User Defined Functions
 Triggers

Home
 Train...Items
 Model...Items
 Train...Items
 Train...Settings

SELECT * FROM c
 Type a query predicate (e.g., WHERE c.id='1'), or choose one from the drop down list, or leave empty to query all documents.

	id	/model_type
☐	model_metrics_2025-07-18T2...	catboost
☐	model_metrics_2025-07-18T2...	catboost
☐	model_metrics_2025-07-18T2...	catboost
☐	model_metrics_2025-07-19T1...	catboost
☐	model_metrics_2025-07-19T1...	catboost
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☐	model_metrics_2025-07-21T0...	catboost
☐	model_metrics_2025-07-21T1...	catboost
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☐	model_metrics_2025-07-21T1...	catboost
☐	model_metrics_2025-07-21T2...	catboost
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☐	model_metrics_2025-07-22T0...	catboost
☐	model_metrics_2025-07-22T0...	catboost

Load more

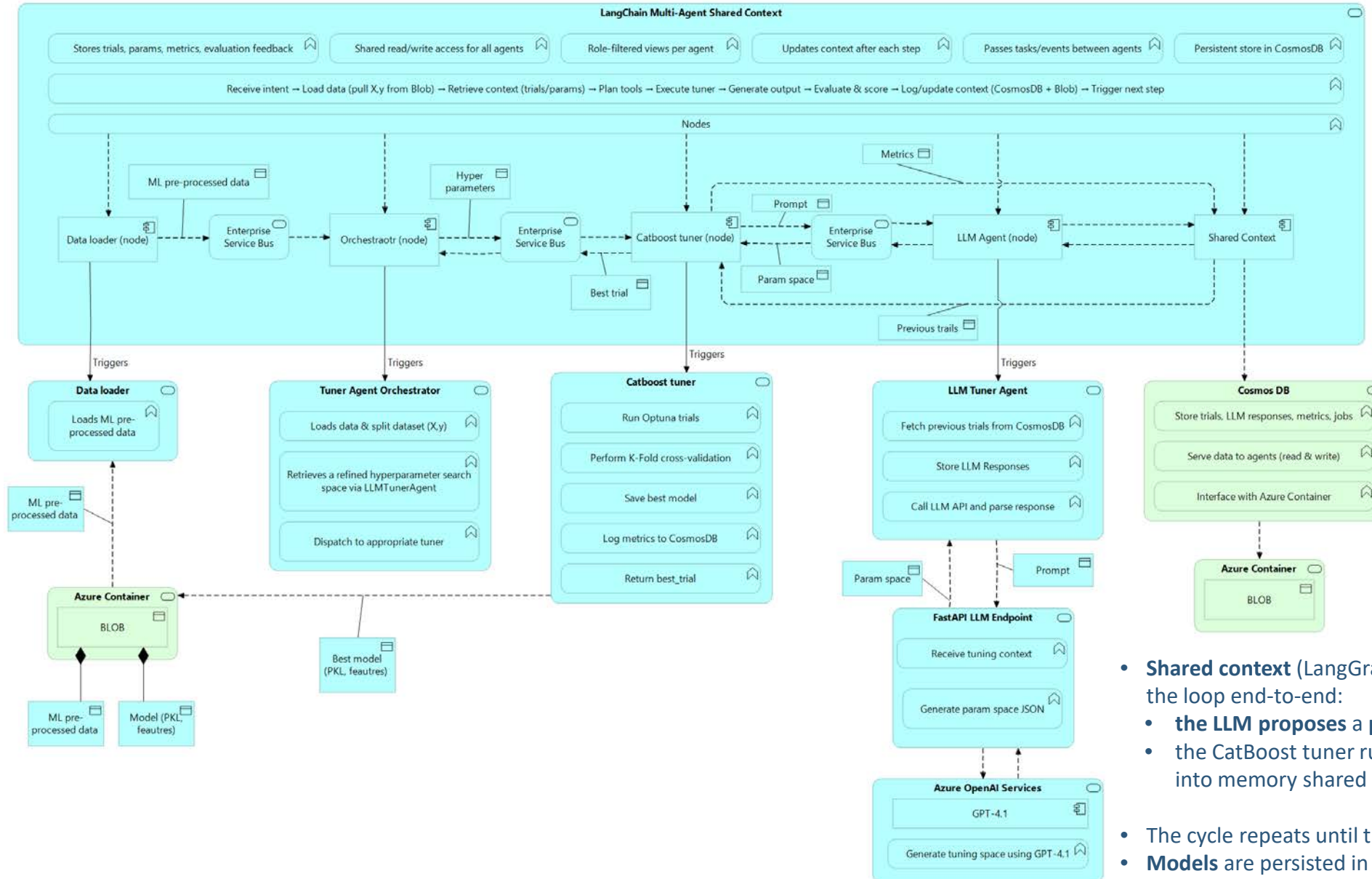
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33  },
34  "feature_importance": [],
35  "training_time": 8702.494880199432,
36  "n_features": 2885,
37  "status": "completed",
38  "is_production_ready": false,
39  "source": "catboost_tuner",
40  "_rid": "MFxFA0SqbJwOAAAAAAAAA==",
41  "_self": "dbs/MFxFAA==/colls/MFxFA0SqbJwOAAAAAAAAA==/",
42  "_etag": "\"2a008aea-0000-4d00-0000-687e54730000\"",
43  "_attachments": "attachments/",
44  "_ts": 1753109619
45 }

```

- Runs are logged to **Cosmos DB** as **JSON** (per trial / training job)
- Key fields: **hyperparameters**, CV scores (**R² / MAE / RMSE**), generalization index, overfitting risk, runtime
- Traceability: **status**, timestamps (start/end), **correlation_id**, **prompt_version**, **best_trial** flag
- Artifacts: **model_uri** (Azure Blob), **feature_manifest**, attachments
- Usage: **dashboards** + **auto-ranking** to select the **best model** and **audit decisions**

LLM-Guided Tuning Orchestration with Azure & Agents (with LangChain & Shared Context)



Each agent is **hosted** in a **dedicated container**

Integration is achieved via

- API Management
- Service Bus
- Event Grid

- **Shared context** (LangGraph + Cosmos DB) lets agents coordinate the loop end-to-end:
- **the LLM proposes a parameter search space**
- the CatBoost tuner runs K-fold trials, and metrics flow back into memory shared context
- The cycle repeats until the **best trial is selected**
- **Models** are persisted in Azure Blob

Model Training & metrics

RANK	TIMESTAMP	MODEL	TRAINING TIME	MAE TRAIN	RMSE TRAIN	R ² TRAIN	MAE TEST	RMSE TEST	R ² TEST	R ² GAP	R ² GAP DIAGNOSTIC	GEN. INDEX	GENERALIZATION	OVERFITTING RISK	N FEATURES	RANKING SCORE
1	25/07/2025 07:42	CatBoost CV (All Features)	5h 49m	63.4k€	88.7k€	0.819927	66.7k€	97.4k€	0.782708	0.037219	Excellent	96	Excellent	Low	2885	0.980
2	25/07/2025 01:53	CatBoost CV (All Features)	59m 16s	62.4k€	87.2k€	0.825684	66.2k€	96.8k€	0.785280	0.040404	Excellent	96	Excellent	Low	2885	0.976
3	25/07/2025 18:51	This dashboard is powered by autonomous AI agents that handle: - Training orchestration (looping models with CV) - Auto-evaluation using advanced metrics (R², MAE, RMSE) - Smart ranking based on generalization, overfitting, and readiness Logging & scoring injected into a dynamic leaderboard Thanks to agent-driven automation, model selection becomes: - Faster, by skipping manual scoring - Smarter, with contextual ranking Production-ready, ensuring only the best models move forward ➡ This enables continuous improvement and scalable evaluation for real-world ML deployment														
4	25/07/2025 14:11								0.782835	0.043691	Excellent	96	Excellent	Low	2885	0.967
10	21/07/2025 21:24								0.804527	0.084134	Moderate	92	Good	Moderate	2885	0.880
11	21/07/2025 07:25								0.808552	0.088318	Moderate	91	Good	Moderate	2885	0.874
12	21/07/2025 23:31								0.807287	0.091249	Moderate	91	Good	Moderate	2885	0.867
13	23/07/2025 01:47								0.809059	0.093151	Moderate	91	Good	Moderate	2885	0.865
14	21/07/2025 14:53								0.808556	0.095579	Moderate	90	Good	Moderate	2885	0.857
15	22/07/2025 04:01								0.809981	0.096399	Moderate	90	Good	Moderate	2885	0.857
30	21/07/2025 04:45								0.809443	0.124655	Strong overfitting	88	Good	High	2885	0.696
31	19/07/2025 19:24								0.815352	0.129577	Strong overfitting	87	Moderate	High	2885	0.690
32	22/07/2025 16:23	CatBoost CV (All Features)	2h 57m	38.7k€	51.8k€	0.938562	60.2k€	90.9k€	0.810664	0.127897	Strong overfitting	87	Moderate	High	2885	0.689
33	20/07/2025 05:15	CatBoost CV (All Features)	3h 27m	37.5k€	49.1k€	0.944668	60.6k€	90.9k€	0.810548	0.134120	Strong overfitting	87	Moderate	High	2885	0.676
34	19/07/2025 15:25	CatBoost CV (All Features)	4h 31m	36.1k€	46.6k€	0.950321	60.2k€	90.5k€	0.812101	0.138220	Strong overfitting	86	Moderate	High	2885	0.668
35	20/07/2025 13:19	CatBoost CV (All Features)	4h 39m	35.0k€	45.2k€	0.953106	59.8k€	90.2k€	0.813565	0.139541	Strong overfitting	86	Moderate	High	2885	0.666
36	20/07/2025 08:39	CatBoost CV (All Features)	3h 24m	34.5k€	44.9k€	0.953738	59.8k€	90.3k€	0.812959	0.140779	Strong overfitting	86	Moderate	High	2885	0.663

Prompt Engineering: The Core of LLM Behavior

- **Prompt engineering shapes how LLMs behave**, it's structured programming for GenAI
- It directly impacts accuracy, cost, and trust

Key Components

- **System Prompt:** Defines LLM persona, role, and response boundaries
- **Instructional Framing:** e.g., *"Answer strictly based on provided context"*
- **Fallback Logic:** *"If unsure, say so. Do not guess."*
- **User Prompt:** The user's actual query, enriched via RAG
- **Template Logic:** Use of placeholders like [CONTEXT], [USER_QUERY]
- **Function Calling:** Schema of external tools callable by the LLM (e.g. get_weather(city))

Key Benefits

- **Minimizes hallucinations** by anchoring answers in trusted documents
- **Delivers fresh insights** without retraining the model
- **Bridges LLMs with enterprise knowledge** (PDFs, SharePoint, wikis...)
- **More scalable and cost-efficient** than fine-tuning

Prompt Example for Catboost Optuna hyperparameters tuning

ROLE DEFINITION — Always start by assigning the model a specific persona.
 # This anchors the LLM's behavior and narrows the scope.
 You are an expert in Optuna hyperparameter tuning for CatBoost regression models.
 Your objective is to generate a parameter space optimized for predicting real estate prices in production environments.

OBJECTIVE — Clearly define what the model must optimize for.
 # The objective must balance performance and generalization.
 # Avoid overfitting to the benchmark.

GOAL:
 # - Improve performance over current baseline (R^2 Test = 0.78)
 # - Maintain excellent generalization (R^2 Gap ≤ 0.05)
 # - Preserve diagnostic integrity for production deployment (Ranking Score ≥ 1.00)

CONTEXT — Provide grounding data to make the task real and reduce hallucinations.
 # Note: Always provide recent metric values to guide reasoning.

Current best model:
 # - R^2 Test = 0.78
 # - R^2 Gap = 0.03
 # - Ranking Score = 0.980

EVALUATION CRITERIA — Guide output prioritization with clear thresholds.
 # LLMs follow these constraints more reliably when explicitly listed.
PRODUCTION READINESS RANKING:
 - $\geq 1.00 \rightarrow$ "Production Ready"
 - $0.95-0.99 \rightarrow$ "Near Production"
 - $0.90-0.94 \rightarrow$ "Development"
 - $< 0.90 \rightarrow$ "Prototype"

STRATEGY — Communicate the logic used to select parameter values.
 # These guiding principles are essential in high-stakes domains like real estate.
OPTIMIZATION STRATEGY:
 1. Prioritize production readiness (Ranking Score ≥ 1.00)
 2. Enforce strong generalization (R^2 Gap ≤ 0.05)
 3. Target moderate improvement (R^2 Test ≥ 0.80)
 4. Avoid aggressive search spaces — conservative tuning only
 5. Tailor parameters for real estate price prediction patterns

PARAMETER SPACE DEFINITION — Use controlled, conservative ranges.
 # This maintains explainability, reduces variance, and respects real-world constraints

Required parameters and suggested ranges (include only these in the JSON output):
 - learning_rate: suggest_loguniform(0.01, 0.15)
 - depth: suggest_int(4, 8)
 - iterations: suggest_int(800, 1800)
 - l2_leaf_reg: suggest_loguniform(2.0, 10.0)
 - border_count: suggest_int(64, 200)
 ...

FORMAT INSTRUCTIONS — Enforce JSON output to ensure API compatibility.
 # Avoid markdown, comments, or explanations — output must be clean JSON.

Use:
 # - "low" / "high" for ranges
 # - "choices" for categoricals
 # - "value" for fixed values

OUTPUT FORMAT:

```
{
  "model": "catboost",
  "param_space": {
    "learning_rate": {
      "method": "suggest_loguniform",
      "low": 0.01,
      "high": 0.15
    },
    "depth": {
      "method": "suggest_int",
      "low": 4,
      "high": 8
    },
    ...
  }
}
```

TARGET OUTPUT QUALITY:
 # - R^2 Test between 0.80 and 0.82
 # - R^2 Gap ≤ 0.05
 # - Ranking Score ≥ 1.00 (PRODUCTION READY)

If previous trials are available, summarize them as context:
 # Here are previous tuning trials:
 # {trial_summary}

Maximizing Model Performance through AI-Orchestrated Tuning

Prompt Analysis & Chat

Prompt Analysis & Performance

Current Performance

78.27%

R² Test Score

3.72%

R² Gap

Ranking Score:

0.980

Status:

Good (Production Ready)

Analysis Actions

START AI ANALYSIS

Get personalized improvements for your prompt

QUICK ANALYSIS

GENERATE NEW

Production Targets

R² Score Progress

78.2% / 85%

6.7 points to target

Overfitting Control

3.7% / 8%

Excellent generalization

Ranking Score

0.980 / 1.0

2 points from production ready

Recent Performance Trend

Last 3 experiments

Average R²: 76.6% | Trend: Improving

ANALYZE TREND

How to use:

Click "Start AI Analysis" or use the quick buttons to get personalized prompt improvements. When the AI suggests a new prompt, use the "Apply" button to replace your current one.

AI Assistant (2 messages)

EXPERT REVIEW:

- Use feature importance and error analysis to guide future preprocessing or feature engineering.
- Log R² gap and resource usage for each trial for traceability.
- Favor CatBoost encoding for high-cardinality categoricals, but check for leakage.
- Outlier removal can improve generalization if outliers are not representative.
- Keep hyperparameter ranges narrow for efficiency unless trial feedback suggests otherwise.

Prompt Example:

CONTEXT:

You are optimizing a CatBoost regression model for Belgian real estate price prediction. The goal is a production-ready solution: R² ≥ 0.85, R² gap ≤ 0.10, interpretable, and efficient (≤ 30 min training, moderate memory). Avoid data leakage.

OBJECTIVE:

Suggest hyperparameters and preprocessing steps to maximize R², minimize overfitting, and ensure production readiness, adapting based on previous trial feedback.

REQUIRED PARAMETERS:

```

python
{
  "learning_rate": trial.suggest_float("learning_rate", 0.01, 0.15, log=True),
  "depth": trial.suggest_int("depth", 4, 9),
  "l2_leaf_reg": trial.suggest_float("l2_leaf_reg", 3.0, 15.0),
  "subsample": trial.suggest_float("subsample", 0.7, 1.0),
  "bagging_temperature": trial.suggest_float("bagging_temperature", 0.0, 1.5),
  "iterations": trial.suggest_int("iterations", 500, 2000),
  "early_stopping_rounds": trial.suggest_int("early_stopping_rounds", 50, 150),
  "border_count": trial.suggest_int("border_count", 64, 180),
  "feature_border_type": trial.suggest_categorical("feature_border_type", ["Median", "GreedyLogSum"]),
  "handle_missing": trial.suggest_categorical("handle_missing", ["MeanImpute", "ZeroImpute", "KNNImpute"]),
  "categorical_encoding": trial.suggest_categorical("categorical_encoding", ["CatBoost", "Ordinal"]),
  "advanced_preprocessing": trial.suggest_categorical("advanced_preprocessing", ["None", "OutlierRemoval", "FeatureInteractions"])
}

```

EVALUATION:

Primary: R² test (≥0.85)
 Secondary: R² gap (≤0.10)
 Ranking: R² test - 0.5 × R² gap

CONSTRAINTS:

Model interpretability, training time ≤ 30 min, no data leakage, practical memory

OPTIMIZATION STRATEGY:

Start conservative, adapt based on feedback, iterate preprocessing, prioritize interpretability and production readiness.

OUTPUT FORMAT:

Return a JSON object with all parameter suggestions and rationale for choices.

This prompt structure will drive efficient, production-ready CatBoost optimization with rigorous controls and actionable, interpretable outputs.

APPLY THIS PROMPT

Click to replace current prompt and start editing (547 chars)

10:10 PM

Ask about the analysis or prompt optimization

SEND

Factor

ESG Agent

Documents RAG

Model Training

Admin Panel

Investment Manager

Manager

Logout

LLM Prompt Agent

Deployment

Current prompt - CATBOOST

v2.1

active

2858 chars

REFRESH

EDIT PROMPT

Current Prompt Content:

COPY PROMPT

EXPORT PROMPT

CONTEXT:

You are optimizing a CatBoost regression model for real estate price prediction using Belgian property data. The objective is to achieve production-level performance.

OBJECTIVE:

Suggest an optimal combination of hyperparameters and preprocessing options to maximize R² score, minimize overfitting, and ensure production readiness.

REQUIRED PARAMETERS TO OPTIMIZE:

Learning Rate Control:
 learning_rate: suggest_float(0.005, 0.1, log=True)

Tree Structure:
 depth: suggest_int(4, 12)
 l2_leaf_reg: suggest_float(1.0, 20.0)

Sampling Strategy:
 subsample: suggest_float(0.5, 1.0)
 bagging_temperature: suggest_float(0.0, 2.0)

Iterations & Early Stopping:
 iterations: suggest_int(500, 2500)
 early_stopping_rounds: suggest_int(50, 250)

Advanced Features:
 border_count: suggest_int(64, 255)
 feature_border_type: suggest_categorical(["Median", "GreedyLogSum", "Uniform"])

Preprocessing:
 handle_missing: suggest_categorical(["MeanImpute", "ZeroImpute"])
 categorical_encoding: suggest_categorical(["OneHot", "Ordinal", "CatBoost"])

EVALUATION CRITERIA:

Primary metric: R² test score (target ≥ 0.85)
 Overfitting control: R² gap ([R² train - R² test], target ≤ 0.10)
 Ranking score: R² test - 0.5 × R² gap

CONSTRAINTS:

Model must be interpretable
 Training time ≤ 30 minutes
 Memory usage must be practical for production deployment

OUTPUT FORMAT:

Return only the following JSON object with all parameter suggestions:

```

python
{

```

Previous Metrics & Results:

REFRESH

Performance Summary

REFRESH

Best Model Performance (Rank #1)

R² Test

0.782708

R² Gap

0.037219

Gap Diagnostic:

Excellent

Gen. Index:

96

Generalization:

Excellent

Overfitting Risk:

Low

Ranking Score

0.980

Production Ready:

Near Production

Features: 2855

Status: Near Production (Good, minor tuning needed)

Recent Results

#1 #96

R² Test: 0.782708

Gap Diagnostic: Excellent

Overfitting Risk: Low

7/25/2025, 7:42:30 AM

R² Gap: 0.037219

Generalization: Excellent

Ranking Score: 0.980

Near Production

#2 #94

R² Test: 0.785000

Gap Diagnostic: Good

Overfitting Risk: Medium

7/25/2025, 5:58:55 AM

R² Gap: 0.047000

Generalization: Good

Ranking Score: 0.971

Development

#3 #95

R² Test: 0.781000

Gap Diagnostic: Good

Overfitting Risk: Low

7/25/2025, 6:51:27 AM

R² Gap: 0.042000

Generalization: Good

Ranking Score: 0.968

Development

Phase 3 – Enterprise-Scale AI Ecosystem

Operate a governed, extensible AI environment at enterprise scale.

8. Security & Compliance in Context-Aware GenAI Architectures

Identity & Access Control, Data Governance & GDPR Compliance, Infrastructure Security



Security & Compliance in Context-Aware GenAI Architectures

Identity & Access Control

- Enforce strict **Role-Based Access Control (RBAC)** for **agent orchestration endpoints (FastAPI, WebSocket)**
- Leverage **Azure Active Directory (AAD)** or **B2C** for **federated identity**
- **Define access scopes per agent** (e.g., ESG Agent visible only to sustainability team)

Data Governance & GDPR Compliance

- **Personal and sensitive data** handled via **context injection (MCP)** must be:
 - Logged with consent
 - **Filtered** via **PII*** **redaction pipelines**
- Ensure data residency and **storage in compliant regions (EU)** using Azure regional policies

Infrastructure Security

- **Isolate agents per container** or Azure Function
- Use **VNet** Integration, **private endpoints**, and **Azure API Management** to protect LLM and **vector services**
- Integrate with **Azure Defender**, **Key Vault**, and **Managed Identity** for runtime secrets and model protection



Security is not optional, it must be baked into every agent interaction

- **PII (Personally Identifiable Information)** = any data that can **identify** a person **directly** or **indirectly** (e.g., name, email, phone, ID, IP, precise location)
- Personal/sensitive data injected via **MCP** must be **logged with explicit consent** and **pseudonymized via PII-redaction** before any LLM call
(ex: [EMAIL_456] about order [ORDER_7782])

9. Monitoring, Tracing & Observability in Azure

*Real-Time Monitoring, Traceability of AI Decisions,
CI/CD + DevOps Integration*



Monitoring, Tracing & Observability in Azure

Real-Time Monitoring

Use **Azure Monitor** and **Application Insights** to:

- **Track latency**, token usage, agent **errors**
- **Visualize** throughput per node (**LangGraph step**)
- **Alert** on API saturation or **service degradation**

Traceability of AI Decisions

- Store all **intermediate steps**:
 - **RAG retrieval inputs/outputs**
 - **Prompt transformations** (via MCP)
 - GPT output and confidence score
- **Store in Azure Cosmos DB** or Data Lake for **auditability**

CI/CD + DevOps Integration

- **Pipelines monitored** with **Azure DevOps**, **GitHub Actions**
- **Rollbacks enabled** on agent model regressions
- Drift detection for **LLM behavior** via **continuous evaluation**

Transparent, traceable agents  Enterprise-grade AI adoption

10. Takeaways: Strategic Impact of AI Agents for the Enterprise

- *What types of strategic agents can be built?*
- *What business value do they deliver?*
- *How to prioritize agent initiatives?*



What types of strategic agents can be built?

- **ESG Agent:**
Automates sustainability reporting and regulatory alignment (CSRD, EU taxonomy...)
- **Tuning Agent:**
Self-optimizes ML workflows via AutoML, **hyperparameter search**, and **performance feedback**
- **Compliance Agent:**
Reviews legal, security and data policies (GDPR, NIS2, ISO27001) across documents or systems
- **Planning Agent:**
Simulates financial or operational scenarios using predictive models and assumptions
- **Change Management Agent:**
Assesses organizational readiness and tracks adoption KPIs for transformation initiatives
- **Executive Briefing Agent:**
Summarizes dashboards, KPIs, and documents for decision-makers in natural language

What business value do they deliver?

- **Reduced cost-to-decision:**
Automates research, retrieval, synthesis and recommendation across knowledge silos
- **Workforce augmentation:**
Delegates complex, repetitive, or knowledge-intensive tasks to intelligent agents
- **Strategic acceleration:**
Enables faster response to regulatory, market or operational signals through real-time insights

Takeaways: Strategic Impact of AI Agents for the Enterprise



CIOs & CDOs should treat **AI Agents** as **modular**, governed **digital co-workers**, embedded within **strategic processes**

LangGraph (agent workflow orchestration): overview & platform docs from LangChain/LangGraph

- <https://www.langchain.com/>
- <https://docs.langchain.com/oss/python/overview>

Model Context Protocol (MCP) for structured tool/context injection across agent

- <https://modelcontextprotocol.io/specification/2025-06-18>
- <https://github.com/modelcontextprotocol>

Vector RAG on Azure (indexing/querying embeddings with Azure AI Search)

- <https://learn.microsoft.com/en-us/azure/search/vector-search-overview>

State & telemetry store using Azure Cosmos DB (NoSQL/vector + JSON)

- <https://learn.microsoft.com/en-us/azure/cosmos-db>

Document source storage via Azure Blob Storage

- <https://learn.microsoft.com/en-us/azure/storage/blobs>

Microsoft Presidio for PII detection/redaction pipelines

- <https://github.com/microsoft/presidio>
- <https://microsoft.github.io/presidio/anonymizer>

THANK YOU

yves.schillings@seccoudis.com
<https://seccoudis.com/>

in



YVES SCHILLINGS



<https://secloudis.com/>

+32 476 42 12 54

in

About Yves

- +15 years supporting CxOs in defining IT strategies aligned with governance, risk, and compliance
- Helps large organizations ensure IT investments support business objectives and deliver value
- Builds bridges between IT and non-IT stakeholders to align technology with marketing, sales, and operations
- Strong experience in pre-sales, bid writing, budgeting, solution design, and business development
- Successfully led complex programs, ensuring delivery excellence and strategic alignment